

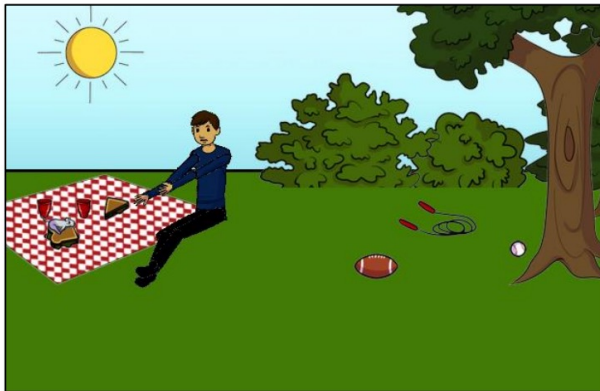
Vision-language models



What color are her eyes?
What is the mustache made of?



How many slices of pizza are there?
Is this a vegetarian pizza?



Is this person expecting company?
What is just under the tree?



Does it appear to be rainy?
Does this person have 20/20 vision?

[Figure source](#)

Outline

- Autoregressive text-to-image generation
- Visual programming
- Visual instruction tuning: LLaVa

Autoregressive text-to-image generation: DALL-E

- Learn a decoder-only transformer model that can be used to generate image based on text prompt



(a) a tapir made of accordion. a tapir with the texture of an accordion.

(b) an illustration of a baby hedgehog in a christmas sweater walking a dog

(c) a neon sign that reads “backprop”. a neon sign that reads “backprop”. backprop neon sign

A. Ramesh et al., [Zero-Shot Text-to-Image Generation](https://openai.com/blog/dall-e/), ICML 2021
<https://openai.com/blog/dall-e/>

DALL-E: Image encoding

- Train convolutional encoder and decoder to compress images to 32x32 grids of discrete tokens (each assuming 8192 values)

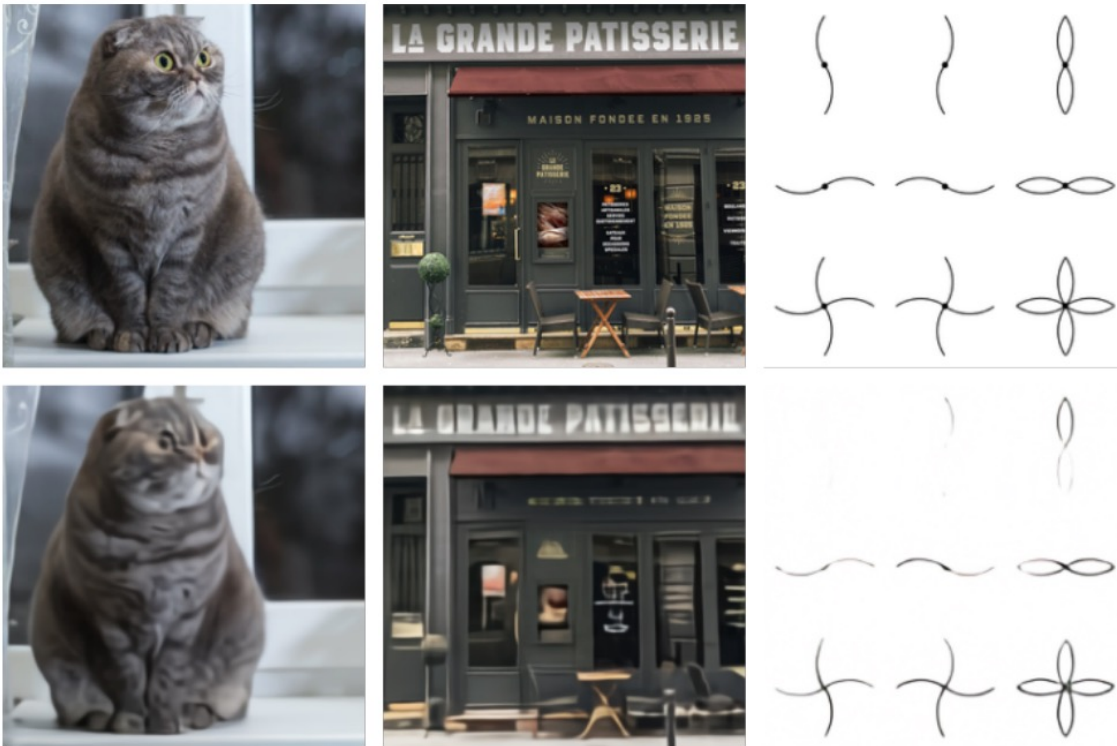


Figure 1. Comparison of original images (top) and reconstructions from the discrete VAE (bottom). The encoder downsamples the spatial resolution by a factor of 8. While details (e.g., the texture of the cat's fur, the writing on the storefront, and the thin lines in the illustration) are sometimes lost or distorted, the main features of the image are still typically recognizable. We use a large vocabulary size of 8192 to mitigate the loss of information.

DALL-E: Transformer architecture and training

- Concatenate up to 256 text tokens with $32 \times 32 = 1024$ image tokens, learn a transformer model with 64 layers and 12B parameters
- Dataset: 250M image-text pairs from the Internet
- Transformer model details
 - Decoder-only architecture
 - 64 self-attention layers,
 - 62 attention heads, sparse attention patterns
 - Mixed-precision training, distributed optimization

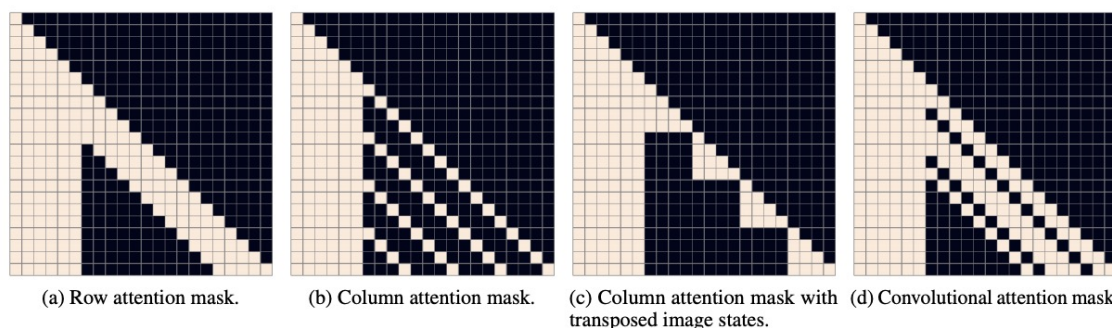


Figure 11. Illustration of the three types of attention masks for a hypothetical version of our transformer with a maximum text length of 6 tokens and image length of 16 tokens (i.e., corresponding to a 4×4 grid). Mask (a) corresponds to row attention in which each image token attends to the previous 5 image tokens in raster order. The extent is chosen to be 5, so that the last token being attended to is the one in the same column of the previous row. To obtain better GPU utilization, we transpose the row and column dimensions of the image states when applying column attention, so that we can use mask (c) instead of mask (b). Mask (d) corresponds to a causal convolutional attention pattern with wraparound behavior (similar to the row attention) and a 3×3 kernel. Our model uses a mask corresponding to an 11×11 kernel.

DALL-E: Generating images given text

- Output samples reranked using CLIP

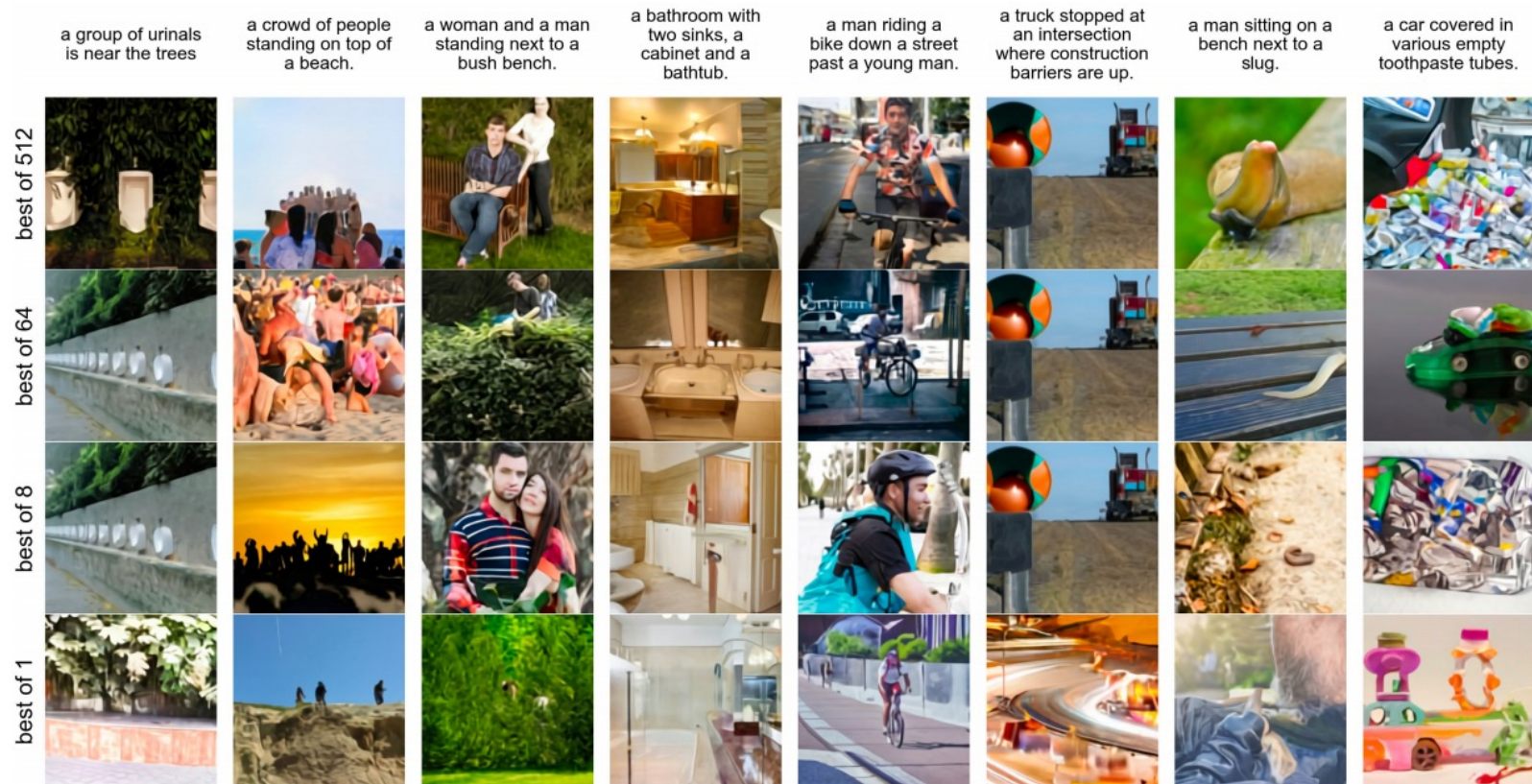
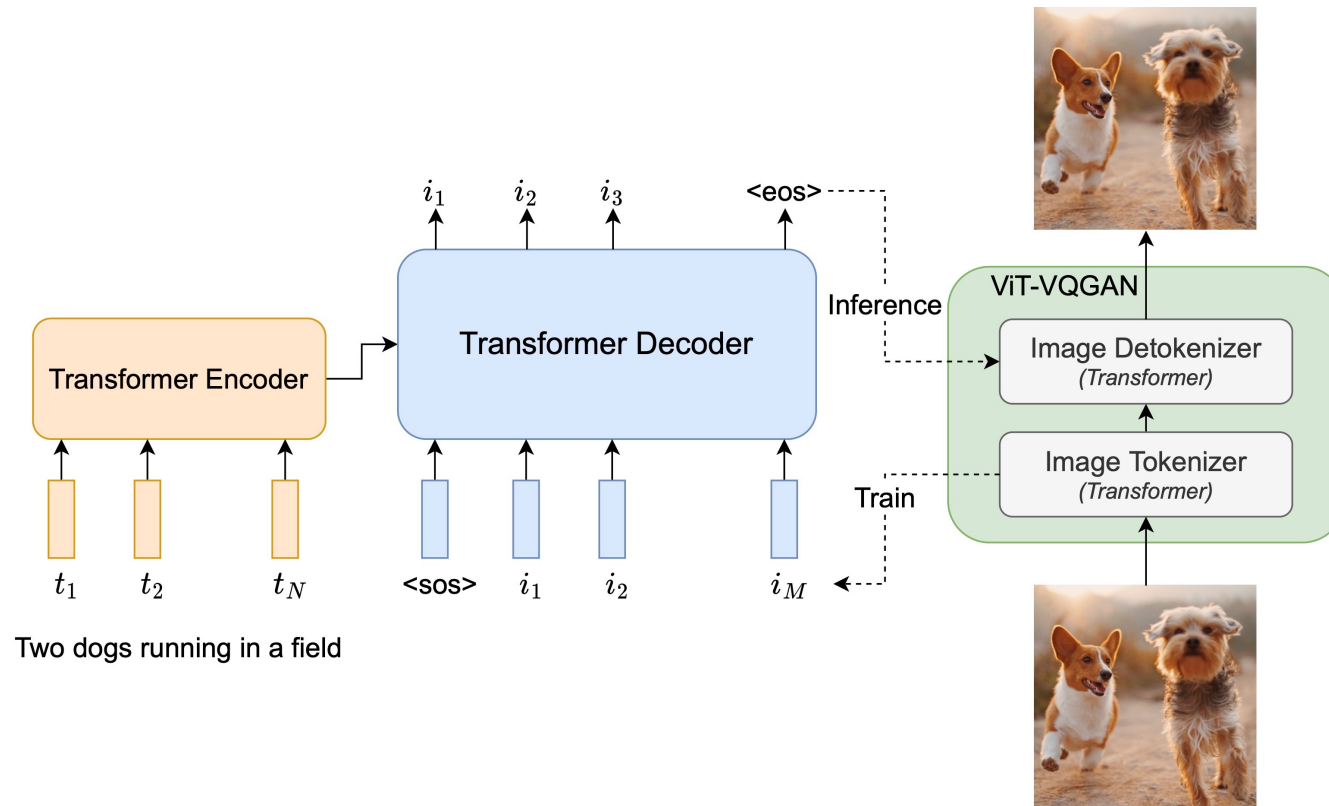


Figure 6. Effect of increasing the number of images for the contrastive reranking procedure on MS-COCO captions.

PARTI: Pathways auto-regressive text-to-image



J. Yu et al. [Scaling Autoregressive Models for Content-Rich Text-to-Image Generation](#). TMLR 2022

PARTI

Model	Encoder Layers	Decoder Layers	Model Dims	MLP Dims	Heads	Total Params
Parti-350M	12	12	1024	4096	16	350M
Parti-750M	12	36	1024	4096	16	750M
Parti-3B	12	36	2048	8192	32	3B
Parti	16	64	4096	16384	64	20B

Parti-350M



Parti-750M



Parti-3B



Parti-20B



A portrait photo of a kangaroo wearing an orange hoodie and blue sunglasses standing on the grass in front of the Sydney Opera House holding a sign on the chest that says Welcome Friends!



A green sign that says "Very Deep Learning" and is at the edge of the Grand Canyon. Puffy white clouds are in the sky.

PART I



A. A photo of a frog reading the newspaper named “Tooday” written on it. There is a frog printed on the newspaper too.



B. A portrait of a statue of the Egyptian god Anubis wearing aviator goggles, white t-shirt and leather jacket. The city of Los Angeles is in the background. Hi-res DSLR photograph.



C. A high-contrast photo of a panda riding a horse. The panda is wearing a wizard hat and is reading a book. The horse is standing on a street against a gray concrete wall. Colorful flowers and the word “PEACE” are painted on the wall. Green grass grows from cracks in the street. DSLR photograph. daytime lighting.

PART I



D. A giant cobra snake made from X . $X \in \{\text{"salad", "pancakes", "sushi", "corn"}\}$

E. A wombat sits in a yellow beach chair, while sipping a martini that is on his laptop keyboard. The wombat is wearing a white panama hat and a floral Hawaiian shirt. Out-of-focus palm trees in the background. DSLR photograph. Wide-angle view.

F. The saying "BE EXCELLENT TO EACH OTHER" ..., (a) brick wall and alien (b) driftwood. (c) old wooden boat with reflection. (d) stained glass. (See text for full prompts.)

PARTI

Approach	Model Type	MS-COCO FID ($\downarrow$)		LN-COCO FID ($\downarrow$)	
		Zero-shot	Finetuned	Zero-shot	Finetuned
Random Train Images [10]	-	2.47		-	
Retrieval Baseline	-	17.97	6.82	33.59	16.48
TReCS [46]	GAN	-	-	-	48.70
XMC-GAN [47]	GAN	-	9.33	-	14.12
DALL-E [2]	Autoregressive	~ 28	-	-	-
CogView [3]	Autoregressive	27.1	-	-	-
CogView2 [61]	Autoregressive	24.0	17.7	-	-
GLIDE [11]	Diffusion	12.24	-	-	-
Make-A-Scene [10]	Autoregressive	11.84	7.55	-	-
DALL-E 2 [12]	Diffusion	10.39	-	-	-
Imagen [13]	Diffusion	7.27	-	-	-
Parti	Autoregressive	7.23	3.22	15.97	8.39

Table 5: Comparison with previous work on the MS-COCO (2014) [16] and Localized Narratives (COCO split) [29] validation sets. When available, we report results for both zero-shot and finetuned models. Retrieval models either perform retrieval over our training set (“zero-shot”), or the respective MS-COCO and LN-COCO training sets (“finetuned”). Parti samples 16 images per text prompt and uses a CoCa model to rank the outputs (Section 2.4). Similar to DALL-E 2 [12], we use guidance scale 1.2 for all above results. We report zero-shot FID score of other model sizes in Figure 9.

Algorithms or data?

- Different image generation methods circa 2023:

Diffusion-based



Auto-regressive



GAN-based

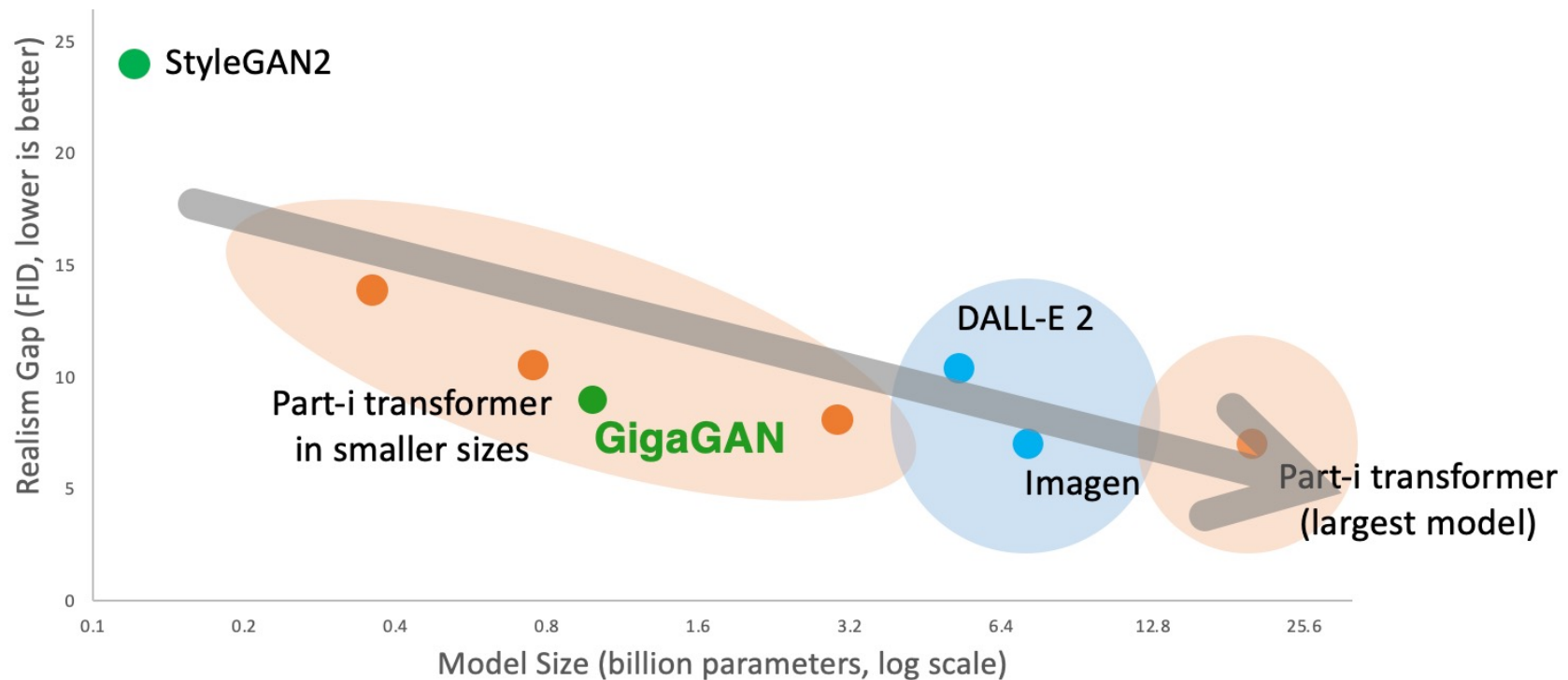


Prompt: *“squirrel reaching for a nut”*

Source: A. Efros

Algorithms or data?

- Image quality vs. model size:



Source: T. Park (via A. Efros)

Outline

- Autoregressive text-to-image generation
- Visual programming

Visual Programming



Figure 1. **VISPROG is a modular and interpretable neuro-symbolic system for compositional visual reasoning.** Given a few examples of natural language instructions and the desired high-level programs, VISPROG generates a program for any new instruction using *in-context learning* in GPT-3 and then executes the program on the input image(s) to obtain the prediction. VISPROG also summarizes the intermediate outputs into an interpretable *visual rationale* (Fig. 4). We demonstrate VISPROG on tasks that require composing a diverse set of modules for image understanding and manipulation, knowledge retrieval, and arithmetic and logical operations.

Visual Programming

- Modules supported in VisProg

Image Understanding	Loc OWL-ViT	FaceDet DSFD (pypi)	Seg MaskFormer	Select CLIP-ViT	Classify CLIP-ViT	Vqa ViLT
	Replace Stable Diffusion	ColorPop PIL.convert() cv2.grabCut()	BgBlur PIL.GaussianBlur() cv2.grabCut()	Tag PIL.rectangle() PIL.text()	Emoji AugLy (pypi)	
Image Manipulation	Crop PIL.crop()	CropLeft PIL.crop()	CropRight PIL.crop()	CropAbove PIL.crop()	CropBelow PIL.crop()	
	List GPT3	Arithmetic & Logical		Eval eval()	Count len()	Result dict()
Knowledge Retrieval						

Visual Programming

- Prompt engineering:

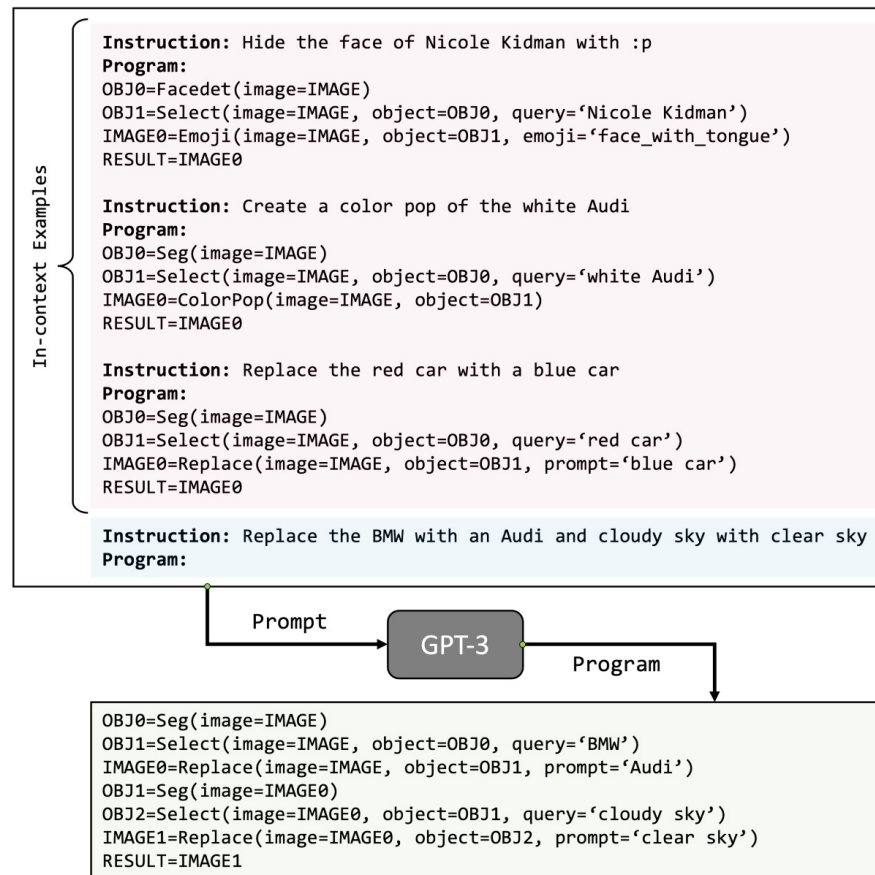
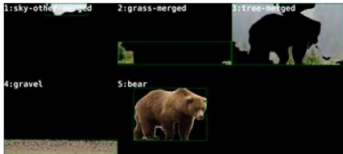


Figure 3. Program generation in VISPROG.

Visual Programming

- Image editing with natural language

Instruction: Replace the ground with white snow and the bear with a white polar bear 		← IMAGE
		← OBJ0=Seg(image=IMAGE)
		← OBJ1=Select(image=IMAGE, object=OBJ0, query='ground')
		← IMAGE0=Replace(image=IMAGE, object=OBJ1, prompt='white snow')
		← OBJ2=Seg(image=IMAGE0)
		← OBJ3=Select(image=IMAGE0, object=OBJ2, query='bear')
Prediction: 		← IMAGE1=Replace(image=IMAGE0, object=OBJ3, prompt='white polar bear')

Visual Programming

- Reasoning about image pairs

Statement: At least three animals are in a flowered field LEFT:  RIGHT:  Prediction: True	 ← LEFT  ← RIGHT 2 ← ANSWER0=Vqa(image=LEFT, question='How many animals are in the flowered field?') 1 ← ANSWER1=Vqa(image=RIGHT, question='How many animals are in the flowered field?') True ← ANSWER2=Eval(expr='{ANSWER0} + {ANSWER1} >= 3?') =Eval(expr='2 + 1 >= 3?')
--	--

Visual Programming

- Example results

Replace Leonardo DiCaprio with Leonardo DiCaprio wearing sunglasses



Create a color pop of the woman in blue and the woman in red and blur the background



Replace Anne Hathaway with Emma Watson and Meryl Streep with Jennifer Lawrence



Replace the desert by sandy beach



Replace the couch with a plush blue couch



Tag the women leaders of Germany, Taiwan, and New Zealand



Tag the three female lead characters from Friends series



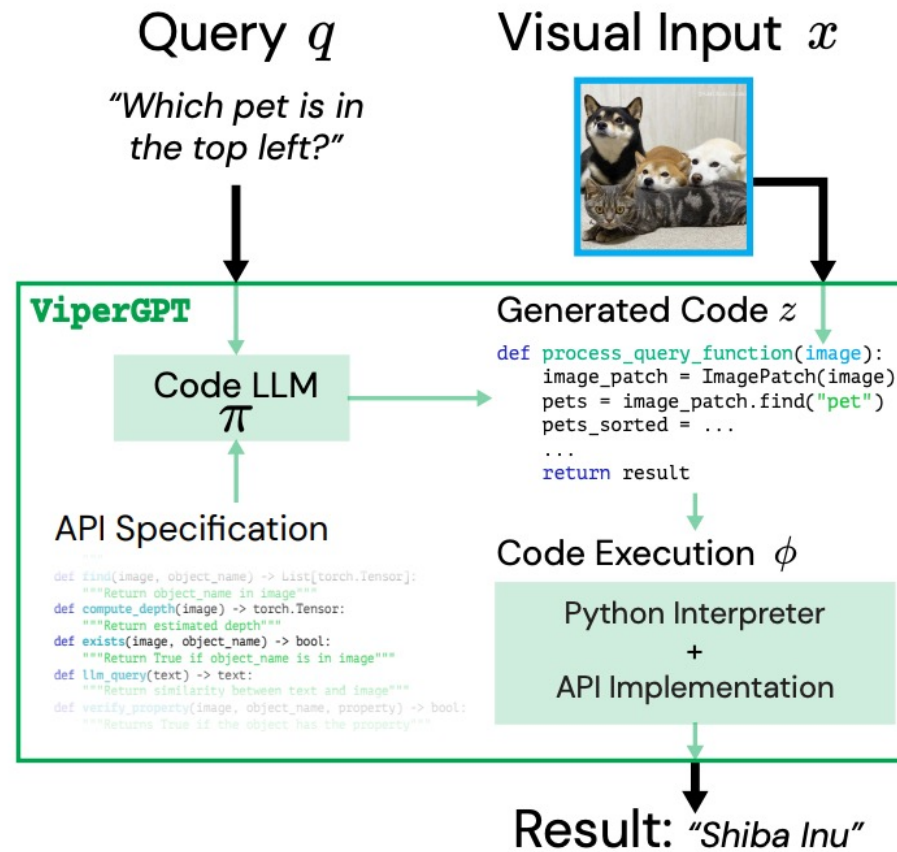
Tag these Scandinavian flags with their countries



Tag the painting of Girl with a Pearl Earring with its painter



ViperGPT



S. Menon et al. [ViperGPT: Visual Inference via Python Execution for Reasoning](#). ICCV 2023

ViperGPT

Query: How many muffins can each kid have for it to be fair?



Generated Code

```
def execute_command(image):  
    image_patch = ImagePatch(image)  
    muffin_patches = image_patch.find("muffin")  
    kid_patches = image_patch.find("kid")  
    return str(len(muffin_patches) // len(kid_patches))
```

Execution

```
muffin_patches =  
image_patch.find("muffin")
```



```
kid_patches =  
image_patch.find("kid")
```



```
► len(muffin_patches)=8  
► len(kid_patches)=2
```

```
► 8//2 = 4
```

Result: 4

Query: Drink with zero alcohol



```
def execute_command(image):  
    image_patch = ImagePatch(image)  
    drink_patches = image_patch.find("drink")  
    for drink_patch in drink_patches:  
        drink_name = drink_patch.simple_query("What is this?")  
        alcoholic = llm_query(f"Does the {drink_name} have alcohol?")  
        if alcoholic == "no":  
            return drink_patch  
    return None
```

► drink_patches=



```
► drink_name = 'tullamore dew'  
► alcoholic = 'yes'
```

```
► drink_name = 'bacardi'  
► alcoholic = 'yes'
```

```
► drink_name = 'gin'  
► alcoholic = 'yes'
```

```
► drink_name = 'dr pepper'  
► alcoholic = 'no'
```

Result:



Query: What would the founder of the brand of the car on the left say to the founder of the brand of the car on the right?

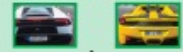


```
def execute_command(image):  
    image_patch = ImagePatch(image)  
    car_patches = image_patch.find("car")  
    car_patches.sort(key=lambda car: car.horizontal_center)  
    left_car = car_patches[0]  
    right_car = car_patches[-1]  
    left_car_brand = left_car.simple_query("What is the brand of this car?")  
    right_car_brand = right_car.simple_query("What is the brand of this car?")  
    left_car_founder = llm_query(f"Who is the founder of {left_car_brand}?")  
    right_car_founder = llm_query(f"Who is the founder of {right_car_brand}?")  
    return llm_query(f"What would {left_car_founder} say to {right_car_founder}?")
```

```
car_patches =  
image_patch.find("car")
```



```
car_patches.sort(...)
```



```
► left_car_brand='lamborghini'  
► right_car_brand='ferrari'
```

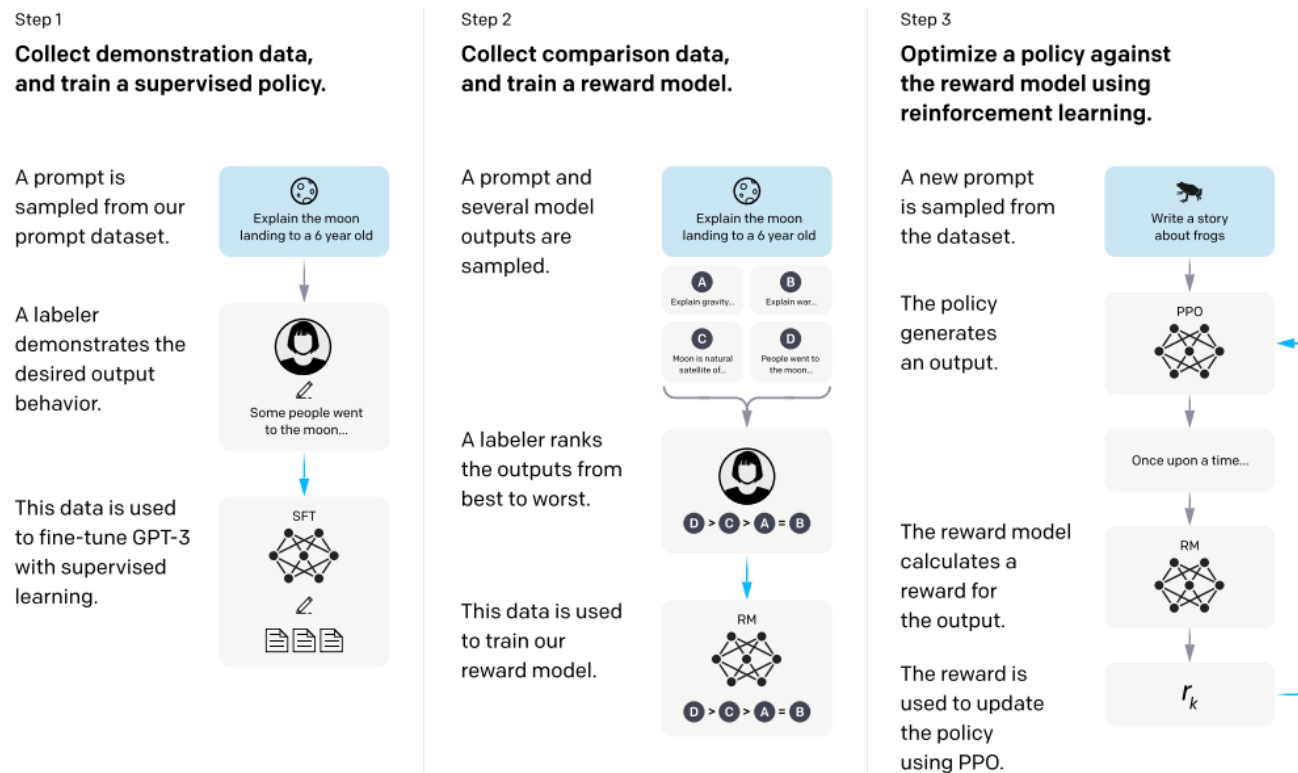
```
► left_car_founder='Ferruccio Lamborghini'  
► right_car_founder='Enzo Ferrari'
```

Result: "Ferruccio Lamborghini might say, 'It's been an honor to be a rival of yours for so many years, Enzo. May our cars continue to push each other to be better and faster!'"

VLMs: Outline

- LLMs: Instruction tuning and chain-of-thought prompting
- Visual instruction tuning
- Current vision-language models
- Trends: unified image and text generation, language diffusion models

InstructGPT: Reinforcement learning with human feedback



L. Ouyang et al. [Training language models to follow instructions with human feedback](#). NeurIPS 2022

InstructGPT: Reinforcement learning with human feedback

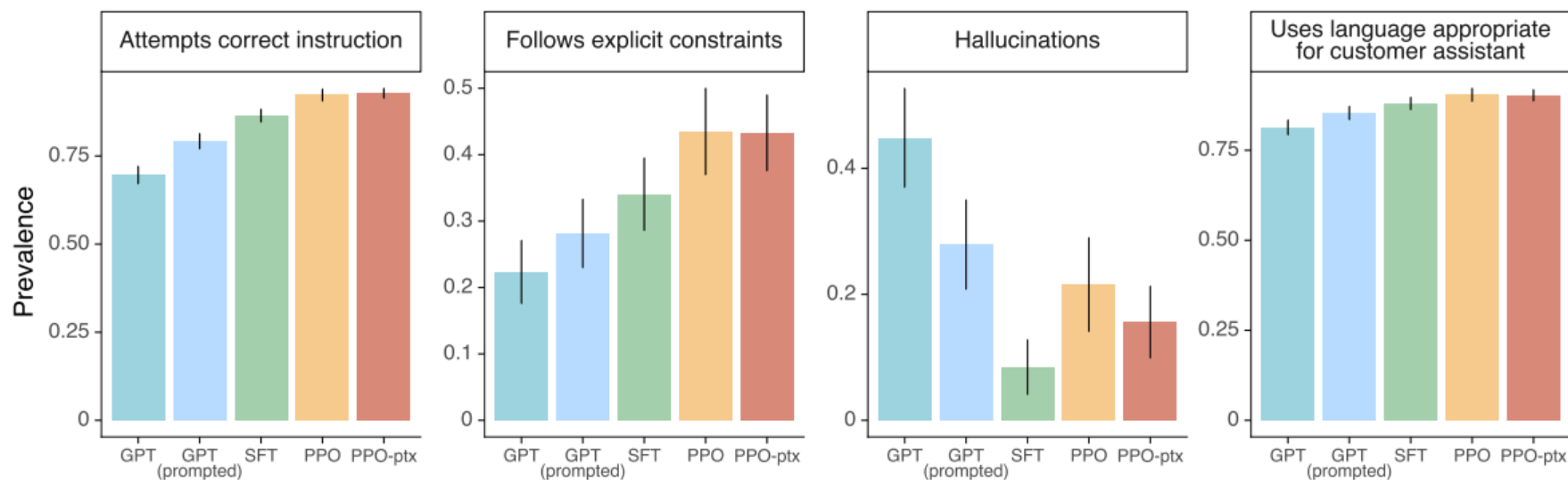


Figure 4: Metadata results on the API distribution. Note that, due to dataset sizes, these results are collapsed across model sizes. See Appendix E.2 for analysis that includes model size. Compared to GPT-3, the PPO models are more appropriate in the context of a customer assistant, are better at following explicit constraints in the instruction and attempting the correct instruction, and less likely to ‘hallucinate’ (meaning, making up information on closed domain tasks like summarization).

L. Ouyang et al. [Training language models to follow instructions with human feedback](#). NeurIPS 2022

RLAIF (Anthropic)

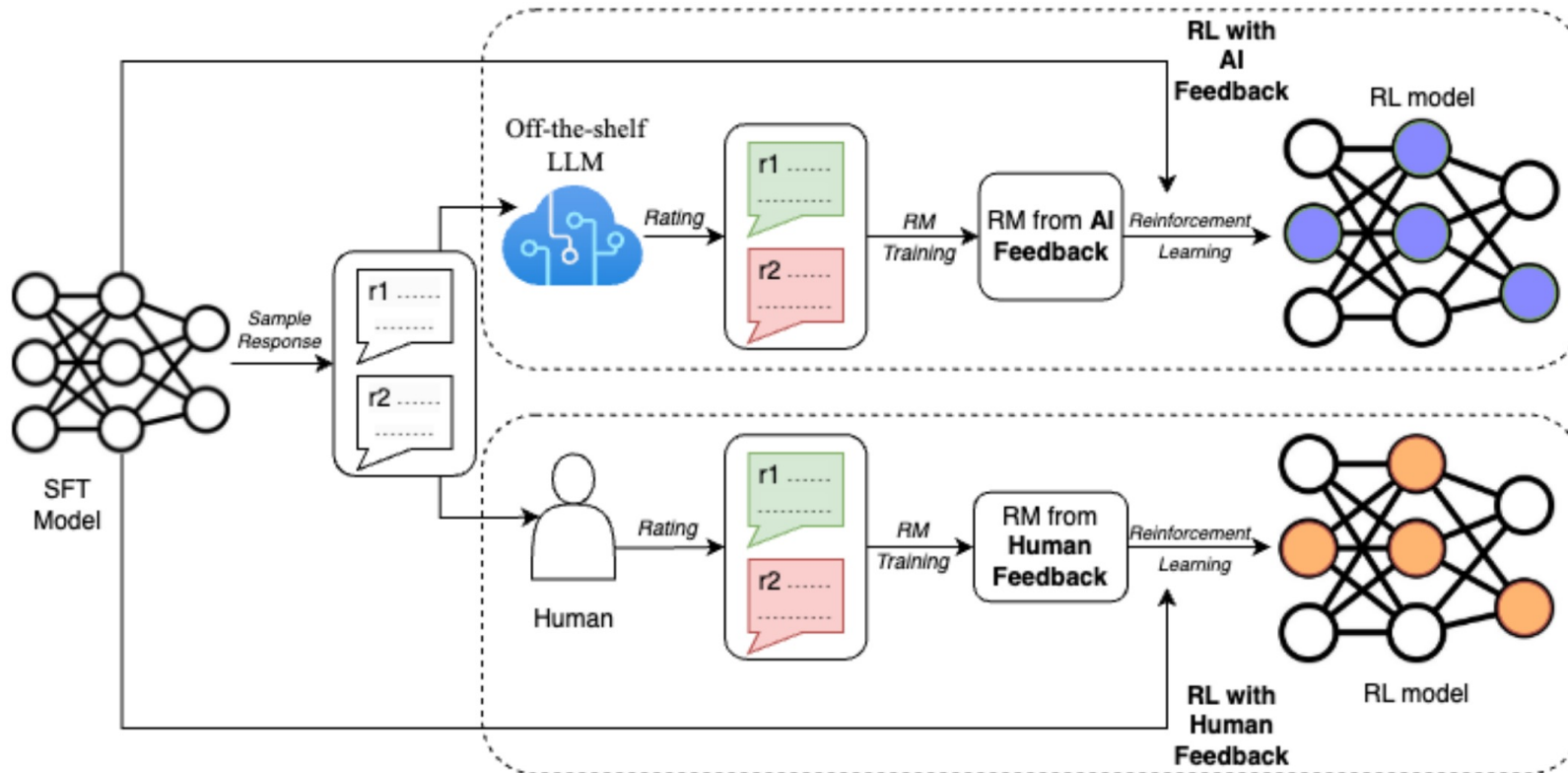
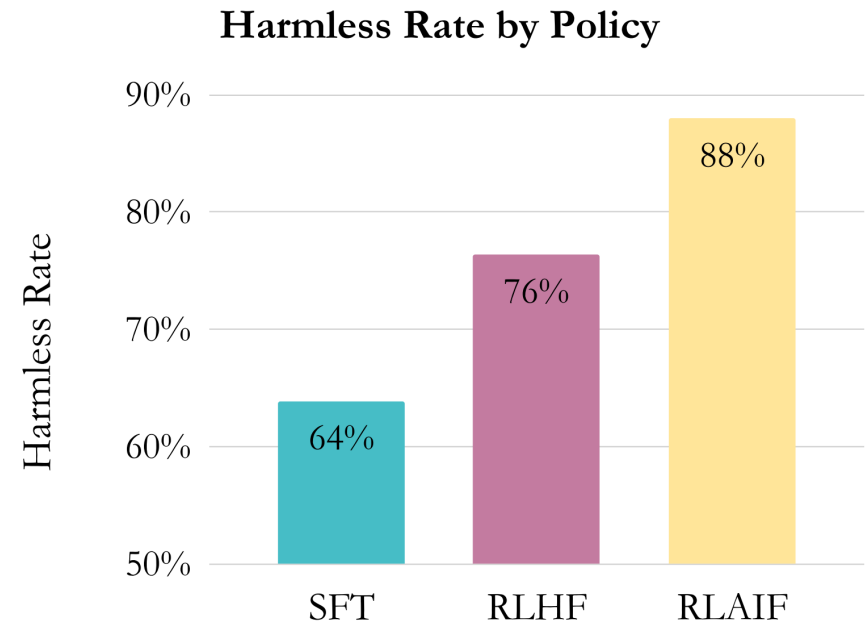
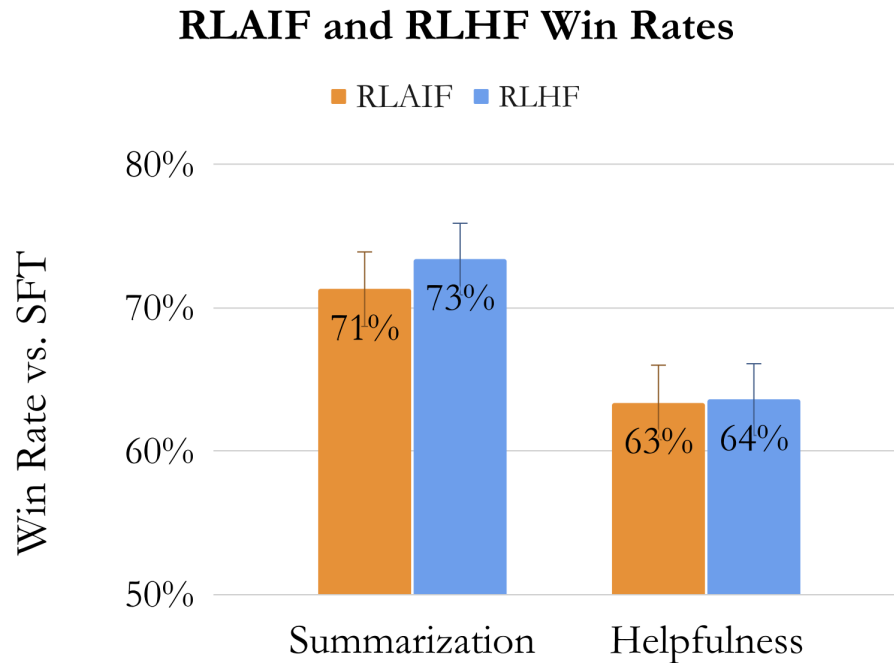


Figure from
Lee et al.

Y. Bai et al. [Constitutional AI: Harmlessness from AI feedback](#). arXiv 2022

H. Lee et al. [RLAIF: Scaling Reinforcement Learning from Human Feedback with AI Feedback](#). arXiv 2023

RLAIF (Anthropic)



From Lee et al.: *Human evaluators strongly prefer RLAIF and RLHF over the SFT baseline for summarization and helpful dialogue generation. Their difference in win rates vs. SFT is not statistically significant. Furthermore, when compared head-to-head, RLAIF is equally preferred to RLHF. For harmless dialogue generation, RLAIF outperforms RLHF.*

Y. Bai et al. [Constitutional AI: Harmlessness from AI feedback](#). arXiv 2022

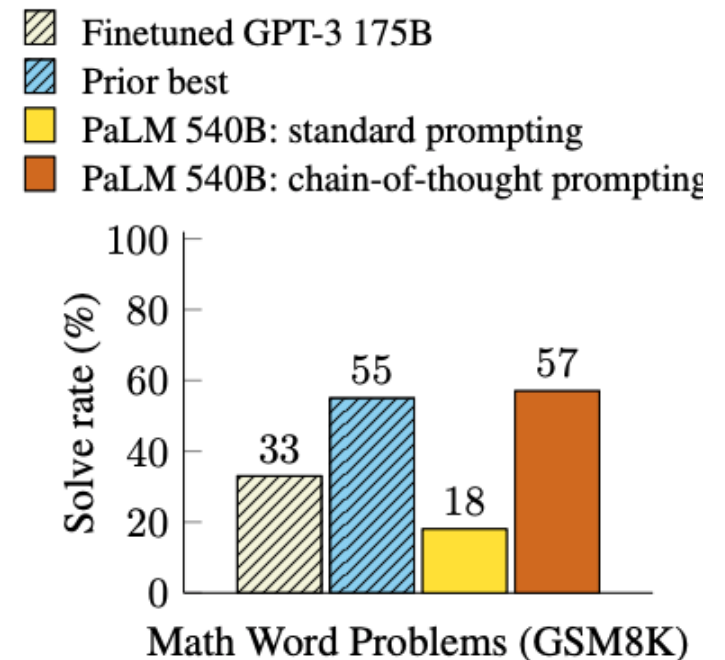
H. Lee et al. [RLAIF: Scaling Reinforcement Learning from Human Feedback with AI Feedback](#). arXiv 2023

Chain-of-thought prompting

Abstract

We explore how generating a *chain of thought*—a series of intermediate reasoning steps—significantly improves the ability of large language models to perform complex reasoning. In particular, we show how such reasoning abilities emerge naturally in sufficiently large language models via a simple method called *chain-of-thought prompting*, where a few chain of thought demonstrations are provided as exemplars in prompting.

Experiments on three large language models show that chain-of-thought prompting improves performance on a range of arithmetic, commonsense, and symbolic reasoning tasks. The empirical gains can be striking. For instance, prompting a PaLM 540B with just eight chain-of-thought exemplars achieves state-of-the-art accuracy on the GSM8K benchmark of math word problems, surpassing even finetuned GPT-3 with a verifier.



Chain-of-thought prompting

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

DeepSeek-R1 (January 2025): RL for CoT

DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

DeepSeek-AI

`research@deepseek.com`

Abstract

We introduce our first-generation reasoning models, DeepSeek-R1-Zero and DeepSeek-R1. DeepSeek-R1-Zero, a model trained via large-scale reinforcement learning (RL) without supervised fine-tuning (SFT) as a preliminary step, demonstrates remarkable reasoning capabilities. Through RL, DeepSeek-R1-Zero naturally emerges with numerous powerful and intriguing reasoning behaviors. However, it encounters challenges such as poor readability, and language mixing. To address these issues and further enhance reasoning performance, we introduce DeepSeek-R1, which incorporates multi-stage training and cold-start data before RL. DeepSeek-R1 achieves performance comparable to OpenAI-o1-1217 on reasoning tasks. To support the research community, we open-source DeepSeek-R1-Zero, DeepSeek-R1, and six dense models (1.5B, 7B, 8B, 14B, 32B, 70B) distilled from DeepSeek-R1 based on Qwen and Llama.

<https://arxiv.org/pdf/2501.12948>

DeepSeek-R1 (January 2025): RL for CoT

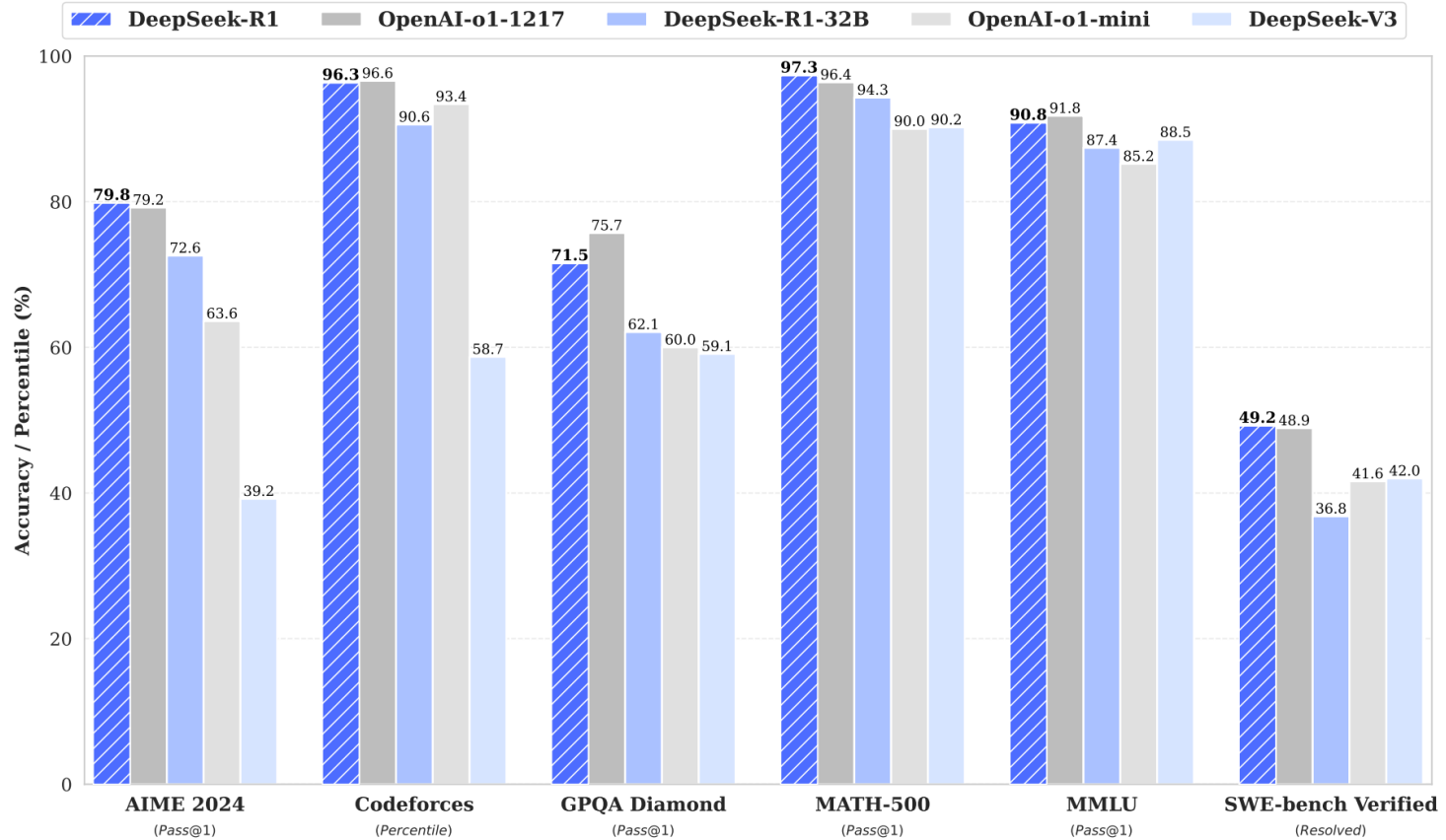


Figure 1 | Benchmark performance of DeepSeek-R1.

Outline

- LLMs: Instruction tuning and chain-of-thought prompting
- Visual instruction tuning: LLaVa

Visual instruction tuning

- Idea: use language-only GPT-4 to generate image-language instruction following data; fine-tune a VLM using this data

Context type 1: Captions

A group of people standing outside of a black vehicle with various luggage.
Luggage surrounds a vehicle in an underground parking area
People try to fit all of their luggage in an SUV.
The sport utility vehicle is parked in the public garage, being packed for a trip
Some people with luggage near a van that is transporting it.



Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...<omitted>

Response type 1: conversation

Question: What type of vehicle is featured in the image?

Answer: The image features a black sport utility vehicle (SUV) ...<omitted>

Response type 2: detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip. ...<omitted>

Response type 3: complex reasoning

Question: What challenges do these people face?

Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...<omitted>

Table 1: One example to illustrate the instruction-following data. The top block shows the contexts such as captions and boxes used to prompt GPT, and the bottom block shows the three types of responses. Note that the visual image is not used to prompt GPT, we only show it here as a reference.

H. Liu et al. [Visual instruction tuning](#).
NeurIPS 2023

Visual instruction tuning

- Example prompt for GPT-4:

```
messages = [ {"role": "system", "content": f"""You are an AI visual assistant, and you are seeing a single image. What you see are provided with five sentences, describing the same image you are looking at. Answer all questions as you are seeing the image.
```

Design a conversation between you and a person asking about this photo. The answers should be in a tone that a visual AI assistant is seeing the image and answering the question. Ask diverse questions and give corresponding answers.

Include questions asking about the visual content of the image, including the **object types, counting the objects, object actions, object locations, relative positions between objects**, etc. Only include questions that have definite answers:

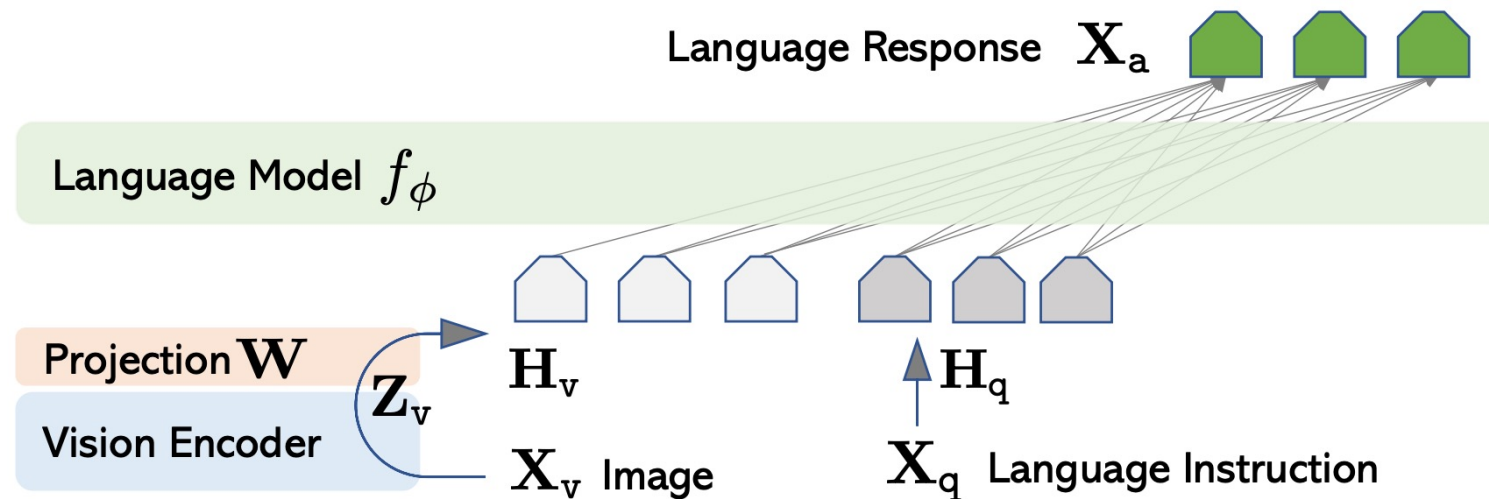
- (1) one can see the content in the image that the question asks about and can answer confidently;
- (2) one can determine confidently from the image that it is not in the image. Do not ask any question that cannot be answered confidently.

Also include complex questions that are relevant to the content in the image, for example, asking about background knowledge of the objects in the image, asking to discuss about events happening in the image, etc. Again, do not ask about uncertain details. Provide detailed answers when answering complex questions. For example, give detailed examples or reasoning steps to make the content more convincing and well-organized. You can include multiple paragraphs if necessary."""}

```
]
for sample in fewshot_samples:
    messages.append({"role": "user", "content": sample['context']})
    messages.append({"role": "assistant", "content": sample['response']})
messages.append({"role": "user", "content": '\n'.join(query)})
```

Visual instruction tuning

- Large Language and Vision Assistant (LLaVa) architecture:



Example LLaVa output



User

What are the meals that I can cook with these?



With the variety of food items stored in the refrigerator, you can prepare several meals. For example, you can create a fruit salad using the strawberries, blueberries, and carrots. Additionally, you can make a delicious smoothie using the fruits and vegetables, which would be a healthy and nutritious snack or breakfast option. The milk and eggs can be used to create various dishes, such as a milk-based smoothie, a baked casserole, or scrambled eggs. The juice can also be used to make a fresh fruit and juice mix or as a refreshing homemade popsicle. The possibilities are vast, and the meals you can create will depend on your preferences and dietary needs.

I would like to do the fruit salad. Show me the recipe.



User



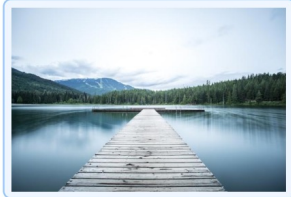
Certainly! Here's a simple fruit salad recipe using the items in your refrigerator:

Ingredients:

- 2 cups fresh strawberries, sliced
- 1 cup fresh blueberries
- 1 large carrot, grated
- Juice of half a lemon
- A handful of chopped fresh parsley or mint
- A few ice cubes
- Optional: honey or maple syrup (adjust to taste)

Example LLaVa output


User



What are the things I should be cautious about when I visit here?


LLaVA

When visiting this peaceful location with a pier extending over a serene lake, surrounded by mountains, there are several things to be cautious about. First, pay attention to the weather conditions, as sudden changes in weather, especially strong winds or storms, can cause the water to become rough and pose risks to the stability of the pier. Second, be mindful of the structural integrity of the pier; avoid overloading it with too many people or excessive weight, as this could potentially damage the pier or cause it to collapse. Additionally, be aware of the surrounding wildlife and respect their space; do not harm or disturb them in any way. Finally, if you decide to go fishing, ensure that you adhere to any local fishing regulations and use appropriate fishing techniques to keep the lake ecosystem healthy and maintain the balance of the food chain.

Compose an engaging travel blog post about my trip to this place, highlighting the beautiful sceneries, joyful experiences, and include possible anecdotes.


LLaVA

Traveling to this picturesque location with a serene lake and mountains in the background was a refreshing and rejuvenating experience. The pier extending over the calm water provided an excellent vantage point from which to appreciate the breathtaking scenery. As I stood on the pier, I couldn't help but feel a deep connection with nature, taking a moment to soak in the peaceful atmosphere.

One of the highlights of my trip was undoubtedly the fishing experience. The area is teeming with life, and I was thrilled to land a sizeable catch, despite the modest size of the lake. The joy of fishing in such a beautiful setting was truly rewarding, and it allowed me to take a break from my busy life and truly immerse myself in the tranquility of nature.

While exploring the surrounding area, I came across a family of birds that had built a nest close to the pier, adding to the charm of the location. Interacting with the wildlife and observing their behavior was an enriching experience that I'll cherish for years to come.

In conclusion, this idyllic destination provided me with an opportunity to unwind and recharge, surrounded by mesmerizing natural beauty and the gentle sounds of water and wildlife. If you're looking for a serene escape from the hustle and bustle of everyday life, this pier overlooking the lake is the perfect place to find solace and reconnect with nature.

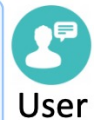
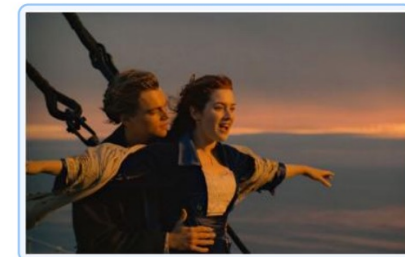

User

Figure 3: LLaVA is capable of recognizing the visual content following the user's intent, without directly prompting for visual recognition. It also provides a detailed response when prompted with a follow-up request, and the generated response is closely related to the provided visual content.

Example LLaVa output



In the image, a young woman is leaning on a man in front of a beautiful sunset, with both of them standing on a boat. They appear to be enjoying a romantic moment together. The scene suggests that they could be depicted as the characters Jack and Rose from the popular movie Titanic. In the movie, their romance blossoms aboard the fictional ocean liner Titanic, with the pair ultimately sharing a passionate kiss as the ship sinks into the ocean. The ending of the movie is known for its emotional and heartfelt scenes, where the characters are forced to say goodbye to each other as they are separated by the sinking ship. The image evokes a sense of romance and nostalgia, paying tribute to the iconic love story portrayed in the movie Titanic.

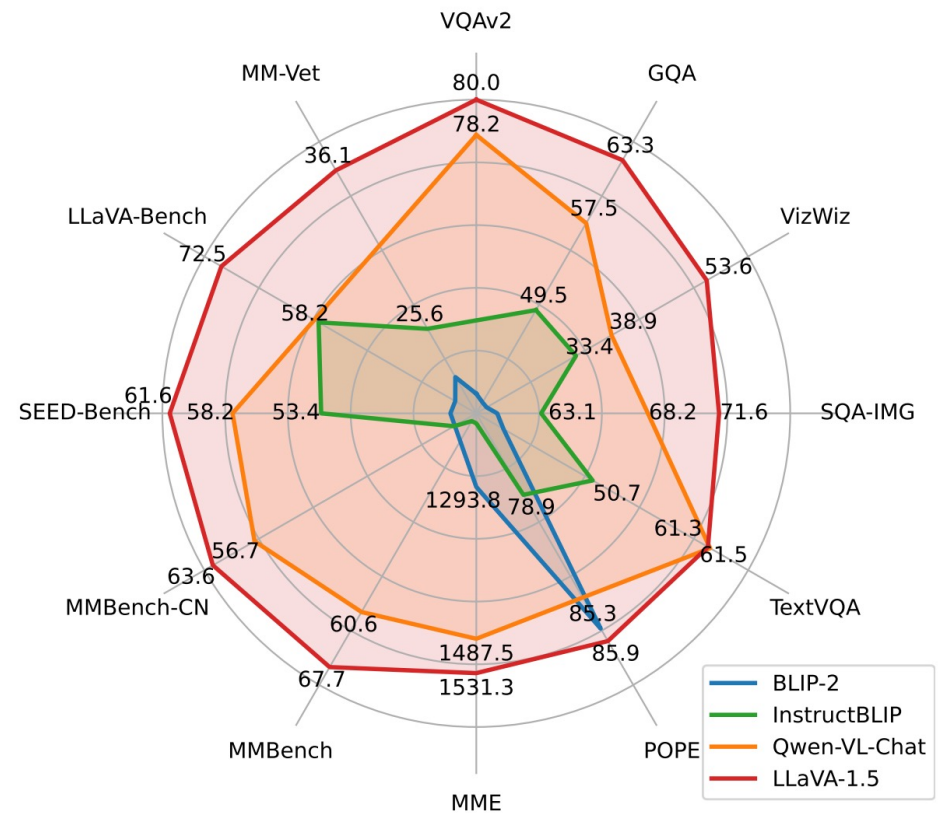
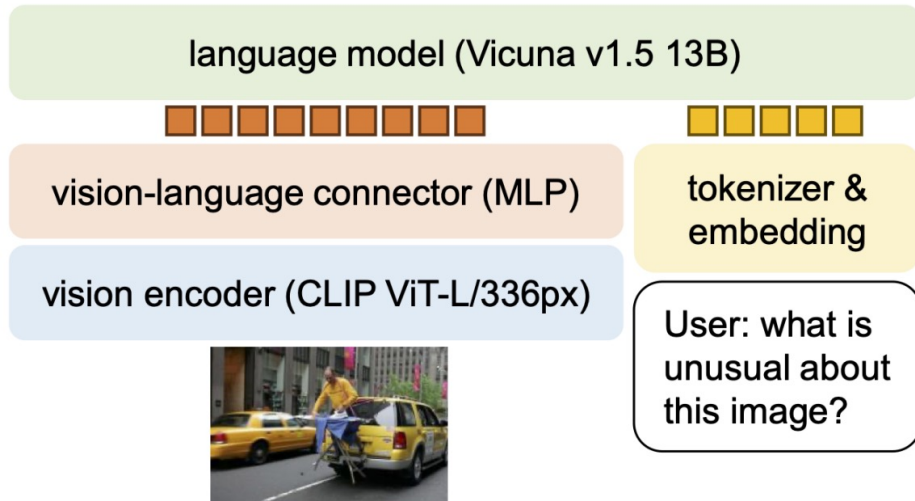


User

What's the ending of this movie?

Figure 4: LLaVA relates the movie scenes to the textual knowledge from the pretrained LLM.

LLaVa-1.5



H. Liu et al. [Improved Baselines with Visual Instruction Tuning](#). CVPR 2024

LLaVa-1.5

Visual input example, Tricky Question:



User	If there are factual errors in the questions, point it out; if not, proceed to answering the question. What's happening in the desert?
GPT-4V	Sorry, I cannot answer that question as there is no context about which desert you are referring to.
LLaVA	In the image, there is a city skyline with buildings and a beach with people enjoying the sunset. The scene is set in a desert, which is unusual for a beach setting.
LLaVA-1.5	There are no deserts in the image. The image features a beach with palm trees, a city skyline, and a large body of water.

Table 5. LLaVA-1.5 can detect and answer tricky questions when prompted to verify the question.

Visual input example, Constrained SD Prompt Generation:



User	Here's a formula for a Stable Diffusion image prompt: a cartoon-style image of [adjective] [subject] [haircut] [clothing] [doing action], [color scheme], detailed, trending on artstation. Generate the prompt for this image.
LLaVA-1.5	A cartoon-style image of a young girl with a bob haircut, wearing a white kimono, holding a stuffed animal, and posing for a picture, with a color scheme of pink and white, detailed, and trending on artstation.

Table 12. Constrained prompt generation for Stable Diffusion. Corresponding components are marked in color.

Outline

- LLMs: Instruction tuning and chain-of-thought prompting
- Visual instruction tuning
- Current vision-language models

Case study of an open-source VLM: InternVL

InternVL: Scaling up Vision Foundation Models and Aligning for Generic Visual-Linguistic Tasks

Zhe Chen^{2,1†}, Jiannan Wu^{3,1†}, Wenhai Wang^{4,1}, Weijie Su^{6,1†}, Guo Chen^{2,1†}, Sen Xing⁵, Muyan Zhong⁵, Qinglong Zhang¹, Xizhou Zhu^{5,7,1}, Lewei Lu^{7,1}, Bin Li⁶, Ping Luo³, Tong Lu², Yu Qiao¹, Jifeng Dai^{5,1✉}

¹OpenGVLab, Shanghai AI Laboratory ²Nanjing University

³The University of Hong Kong ⁴The Chinese University of Hong Kong ⁵Tsinghua University

⁶University of Science and Technology of China ⁷SenseTime Research

<https://github.com/OpenGVLab/InternVL>

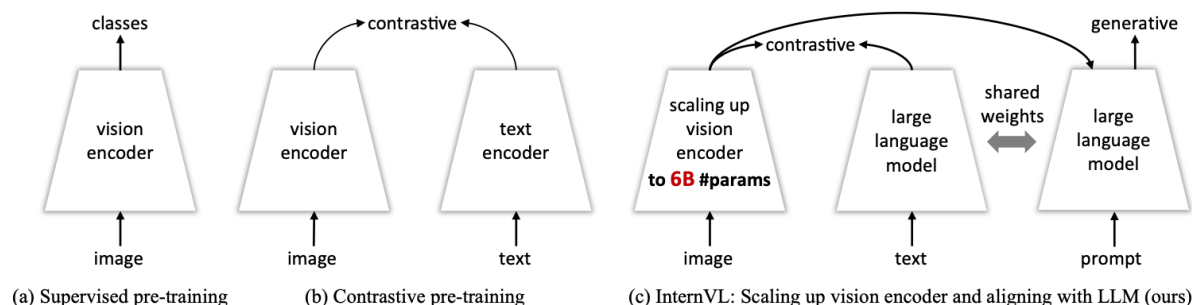


Figure 1. **Comparisons of different vision and vision-language foundation models.** (a) indicates the traditional vision foundation model, *e.g.* ResNet [43] pre-trained on classification tasks. (b) represents the vision-language foundation models, *e.g.* CLIP [89] pre-trained on image-text pairs. (c) is our InternVL, which presents a workable way to align the large-scale vision foundation model (*i.e.*, InternViT-6B) with the large language model and is versatile for both contrastive and generative tasks.

Z. Chen et al. [InternVL: Scaling up Vision Foundation Models and Aligning for Generic Visual-Linguistic Tasks.](#)

CVPR 2024

InternVL

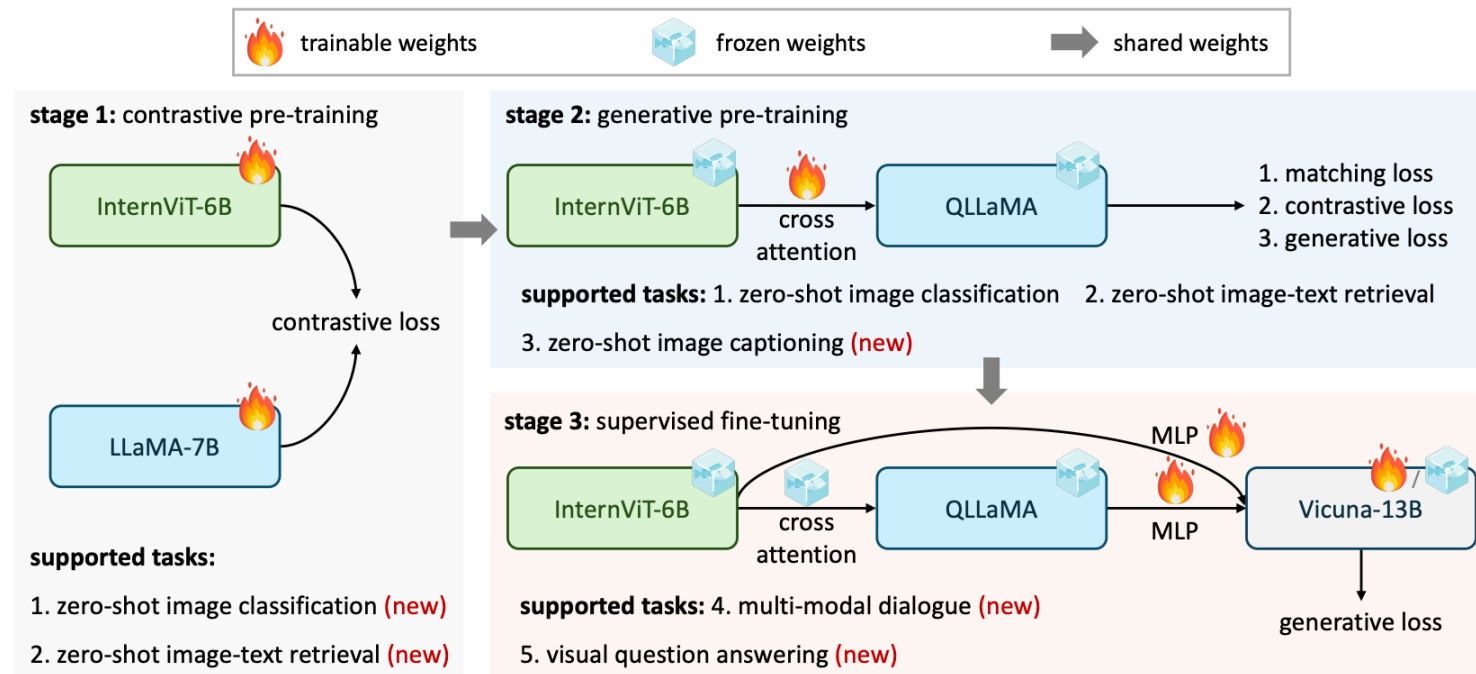


Figure 3. **The training strategy of the proposed InternVL model.** It consists of three progressive stages, including vision-language contrastive training, vision-language generative training, and supervised fine-tuning. These stages effectively leverage public data from diverse sources, ranging from noisy image-text pairs on the web to high-quality caption, VQA, and multi-modal dialogue datasets.

InternVL

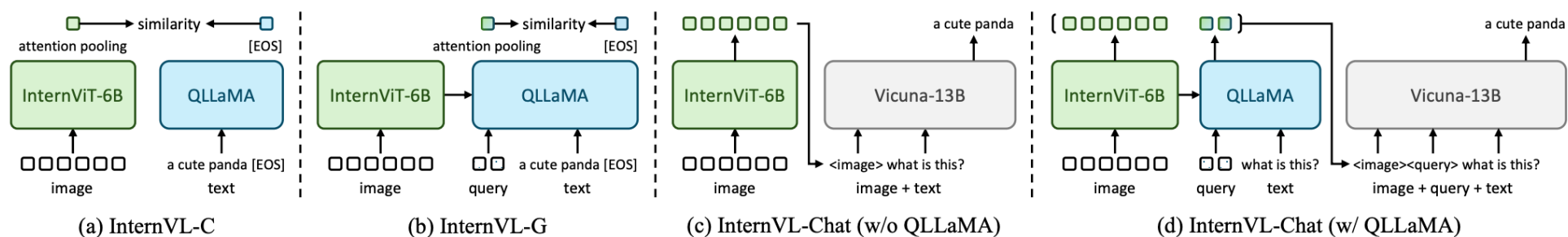


Figure 4. **Different ways to use InternVL.** By flexibly combining the vision encoder and the language middleware, InternVL can support various vision-language tasks, including contrastive tasks, generative tasks, and multi-modal dialogue.

InternVL

dataset	characteristics		stage 1		stage 2	
	language	original	cleaned	remain	cleaned	remain
LAION-en [91]	English	2.3B	1.94B	84.3%	91M	4.0%
LAION-COCO [92]		663M	550M	83.0%	550M	83.0%
COYO [12]		747M	535M	71.6%	200M	26.8%
CC12M [18]		12.4M	11.1M	89.5%	11.1M	89.5%
CC3M [94]		3.0M	2.6M	86.7%	2.6M	86.7%
SBU [85]		1.0M	1.0M	100%	1.0M	100%
Wukong [41]	Chinese	100M	69.4M	69.4%	69.4M	69.4%
LAION-multi [91]	Multi	2.2B	1.87B	85.0%	100M	4.5%
Total	Multi	6.03B	4.98B	82.6%	1.03B	17.0%

Table 2. **Details of the training data for InternVL in stage 1 and stage 2.** Among them, LAION-en [91], LAION-multi [91], COYO [12], and Wukong [41] are web-scale image-text pairs data. LAION-COCO [92] is a synthetic dataset with high-quality captions from LAION-en. CC12M [18], CC3M [94], SBU [85] are academic caption datasets. “Multi” means multilingual.

task	#samples	dataset
Captioning	588K	COCO Caption [20], TextCaps [96]
VQA	1.1M	VQAv2 [40], OKVQA [79], A-OKVQA [93], IconQA [75], AI2D [53], GQA [48]
OCR	294K	OCR-VQA [82], ChartQA [80], DocVQA [25], ST-VQA [11], EST-VQA [116], InfoVQA [81], LLaVAR [143]
Grounding	323K	RefCOCO+/+g [78, 132], Toloka [108]
Grounded Cap.	284K	RefCOCO+/+g [78, 132]
Conversation	1.4M	LLaVA-150K [69], SVIT [144], VisDial [29], LRV-Instruction [67], LLaVA-Mix-665K [68]

Table 3. **Details of the training data for InternVL in stage 3.** We collect a wide range of high-quality instruction data, totaling approximately 4 million samples. For a fair comparison, we only use the training split of these datasets.

InternVL

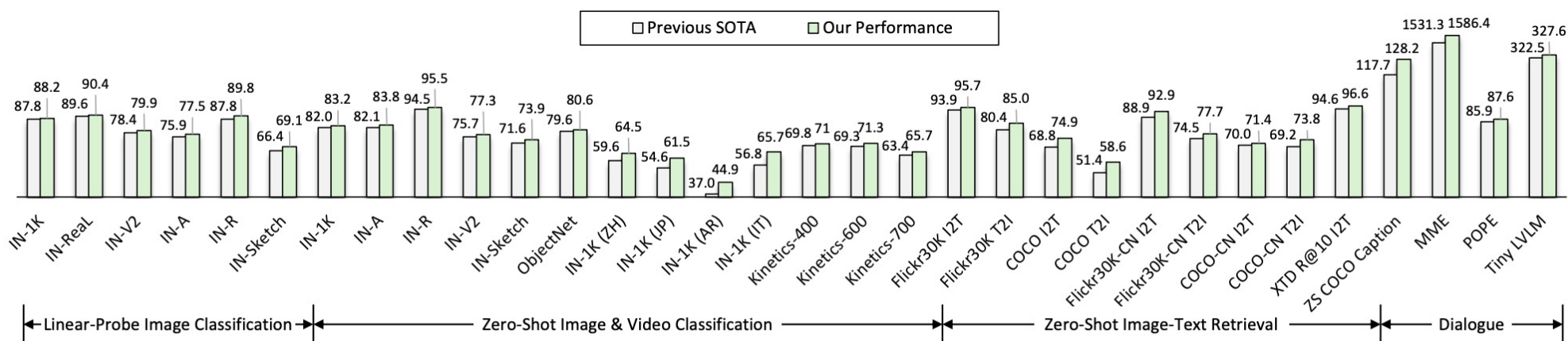


Figure 2. **Comparison results on various generic visual-linguistic tasks**, including image classification, video classification, image-text retrieval, image captioning, and multi-modal dialogue. The proposed InternVL achieves the best performance on all these tasks. Note that only the models trained on public data are included. “IN” is an abbreviation for ImageNet [31].

InternVL 2.5

Expanding Performance Boundaries of Open-Source Multimodal Models with Model, Data, and Test-Time Scaling

Zhe Chen^{4,1*†}, Weiyun Wang^{5,1*†}, Yue Cao^{4,1*†}, Yangzhou Liu^{4,1*†}, Zhangwei Gao^{7,1*†}, Erfei Cui^{7,1*†}, Jinguo Zhu^{1*}, Shenglong Ye^{1*}, Hao Tian^{2*}, Zhaoyang Liu^{1*†}, Lixin Gu¹, Xuehui Wang^{1†}, Qingyun Li^{1†}, Yimin Ren^{1†}, Zixuan Chen², Jiapeng Luo², Jiahao Wang², Tan Jiang², Bo Wang², Conghui He¹, Botian Shi¹, Xingcheng Zhang¹, Han Lv¹, Yi Wang¹, Wenqi Shao¹, Pei Chu¹, Zhongying Tu¹, Tong He¹, Zhiyong Wu¹, Huipeng Deng¹, Jiaye Ge¹, Kai Chen¹, Kaipeng Zhang¹, Limin Wang^{4,1}, Min Dou¹, Lewei Lu², Xizhou Zhu^{3,1}, Tong Lu⁴, Dahua Lin^{6,1}, Yu Qiao¹, Jifeng Dai^{3,1✉}, Wenhai Wang^{6,1✉}

¹Shanghai AI Laboratory ²SenseTime Research ³Tsinghua University ⁴Nanjing University
⁵Fudan University ⁶The Chinese University of Hong Kong ⁷Shanghai Jiao Tong University

Code: <https://github.com/OpenGVLab/InternVL>

Model: https://huggingface.co/OpenGVLab/InternVL2_5-78B

HF Demo: <https://huggingface.co/spaces/OpenGVLab/InternVL>

Z. Chen et al. [Expanding Performance Boundaries of Open-Source Multimodal Models with Model, Data, and Test-Time Scaling](#). arXiv 2025

InternVL 2.5

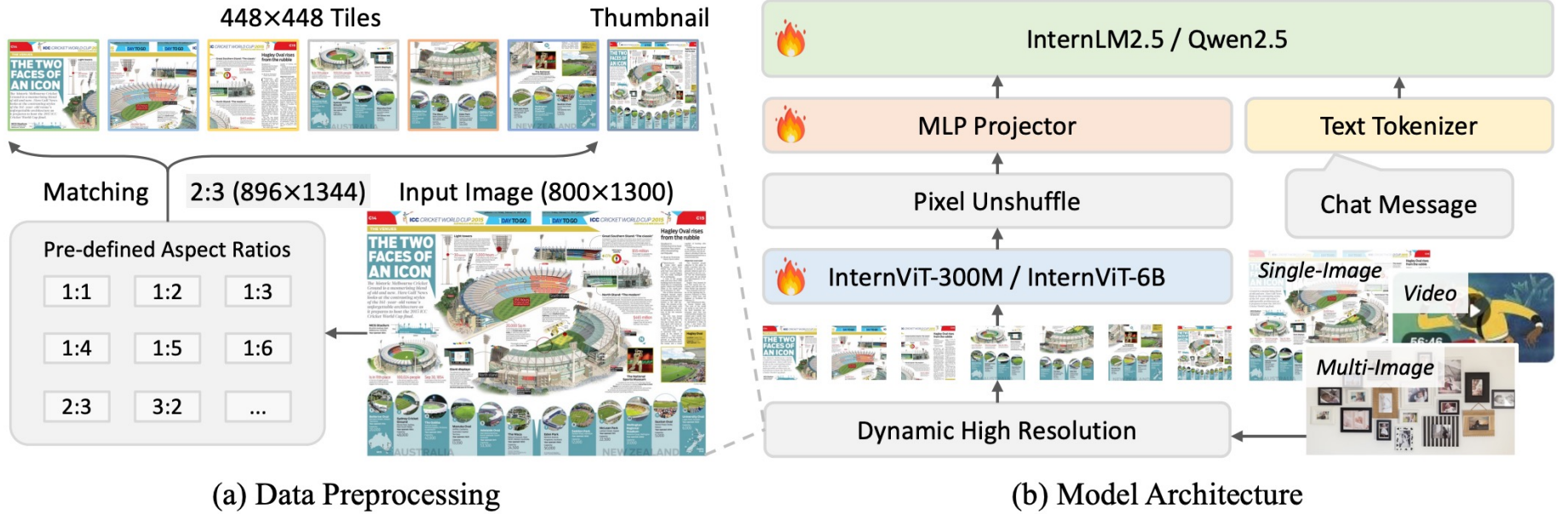


Figure 2: Overall architecture. InternVL 2.5 retains the same model architecture as InternVL 1.5 [35] and InternVL 2.0, *i.e.* the widely-used “ViT-MLP-LLM” paradigm, which combines a pre-trained InternViT-300M or InternViT-6B with LLMs [19, 229] of various sizes via an MLP projector. Consistent with previous versions, we apply a pixel unshuffle operation to reduce the 1024 visual tokens produced by each 448×448 image tile to 256 tokens. Moreover, compared to InternVL 1.5, InternVL 2.0 and 2.5 introduced additional data types, incorporating multi-image and video data alongside the existing single-image and text-only data.

InternVL 2.5

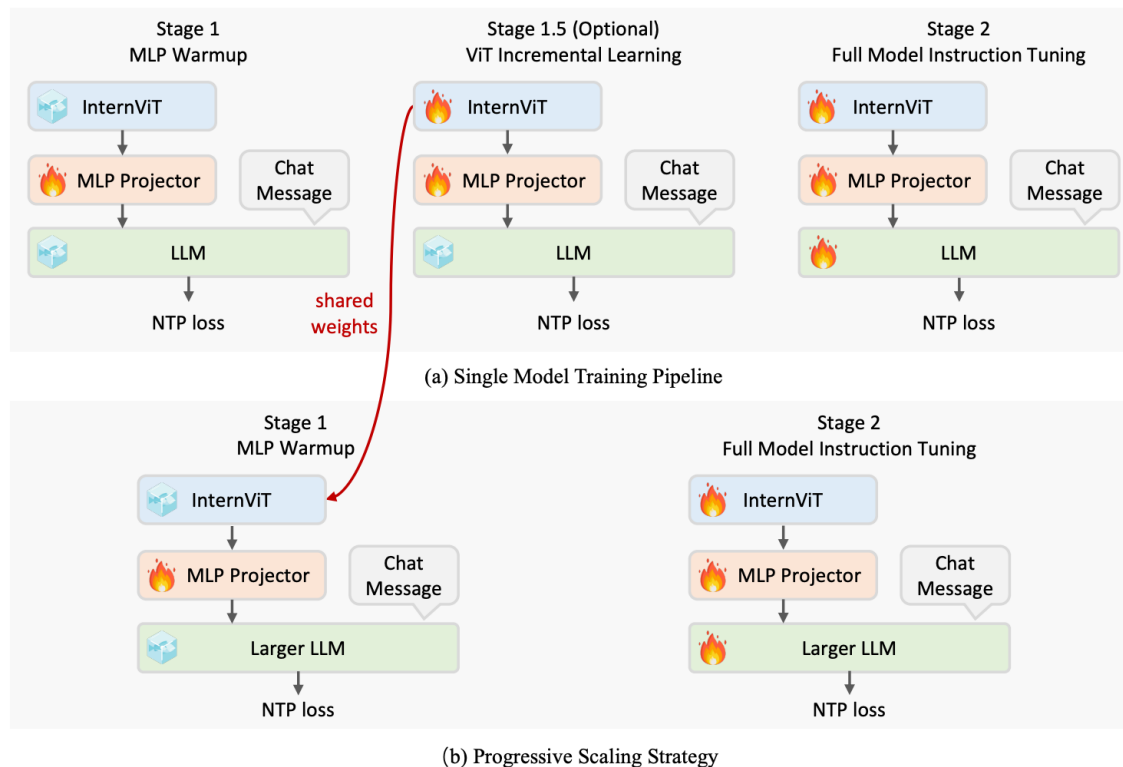


Figure 4: **Illustration of the training pipeline and progressive scaling strategy.** (a) Single model training pipeline. The training process is divided into three stages—Stage 1 (MLP warmup), optional Stage 1.5 (ViT incremental learning), and Stage 2 (full model instruction tuning). The multi-stage design progressively enhances vision-language alignment, stabilizes training, and prepares modules for integration with larger LLMs. (b) Progressive scaling strategy. The ViT module trained with a smaller LLM in earlier stages can be easily integrated with larger LLMs, enabling scalable model alignment with affordable resource overhead.

InternVL 2.5

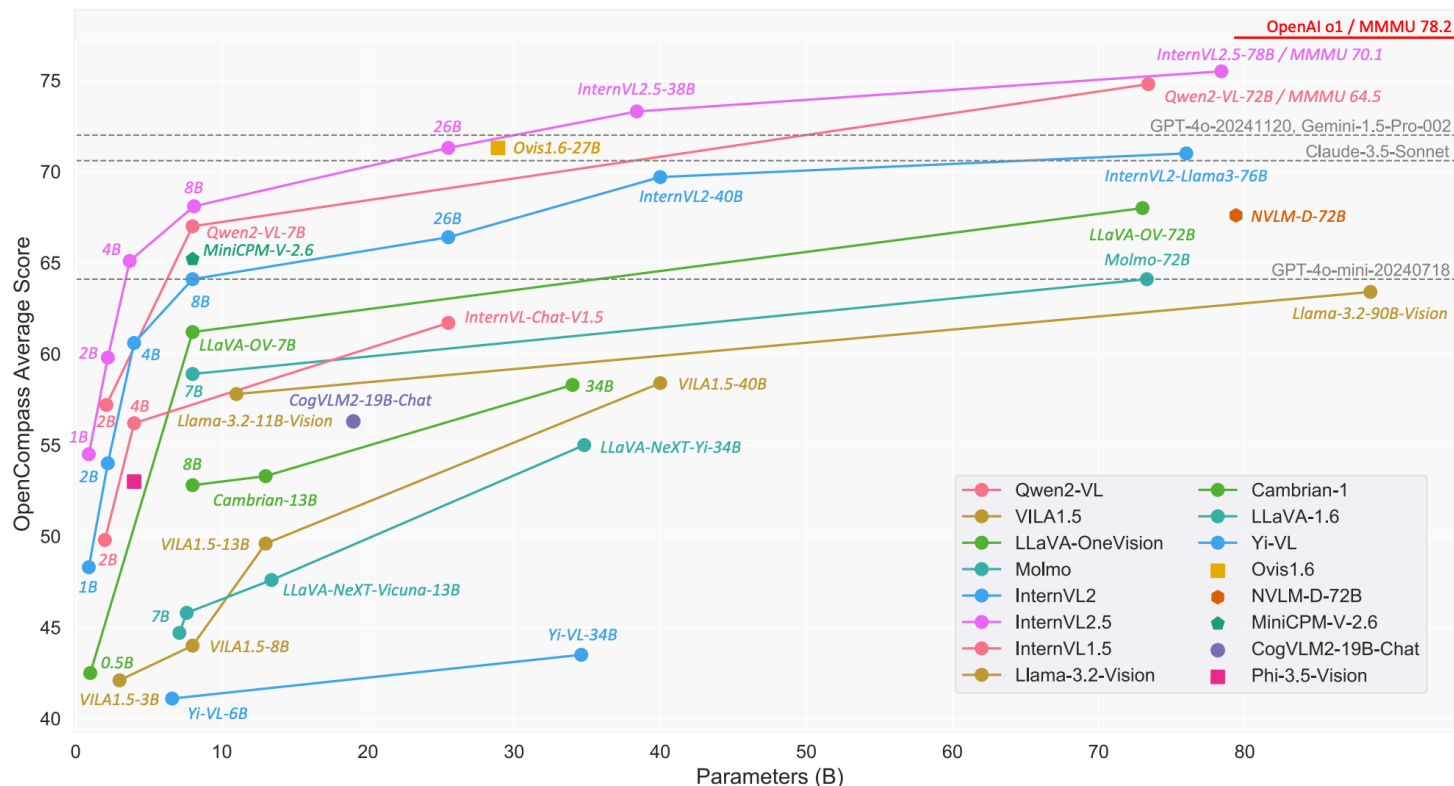


Figure 1: **Performance of various MLLMs on the OpenCompass leaderboard.** InternVL 2.5 showcases strong multimodal capabilities, rivaling closed-source models like GPT-4o [192] and Claude-3.5-Sonnet [8]. However, since the OpenCompass score is derived from 8 academic VQA benchmarks and covers only a subset of overall capabilities, we still need further effort to match the performance with closed-source models.

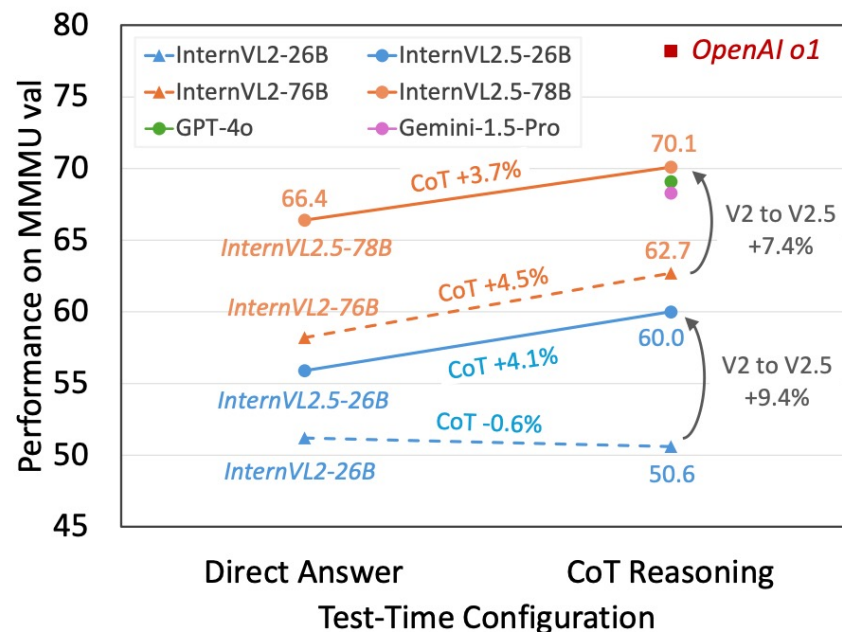
CoT in InternVL 2.5

Answer the preceding multiple choice question. The last line of your response should follow this format: 'Answer: \boxed{\$LETTER}' (without quotes), where LETTER is one of the options. If you are uncertain or the problem is too complex, make a reasoned guess based on the information provided. Avoid repeating steps indefinitely—provide your best guess even if unsure. Think step by step logically, considering all relevant information before answering.

(a) CoT prompt for multiple-choice questions

Answer the preceding question. The last line of your response should follow this format: 'Answer: \boxed{\$FINAL_ANSWER}' (without quotes), where 'FINAL_ANSWER' is your conclusion based on the reasoning provided. If you are uncertain or the problem is too complex, make a reasoned guess based on the information provided. Avoid repeating steps indefinitely—provide your best guess even if unsure. Think step by step logically, considering all relevant information before answering.

(b) CoT prompt for open-ended questions



(c) Model performance on MMMU val

Figure 9: CoT prompts used in our model testing. By leveraging these prompts for CoT reasoning, we can scale up testing time, significantly enhancing the performance of InternVL 2.5 models on MMMU [289].

Outline

- LLMs: Instruction tuning and chain-of-thought prompting
- Visual instruction tuning
- Current vision-language models
- Trends: unified image and text generation, language diffusion models

Unified image and text generation: Chameleon

Chameleon: Mixed-Modal Early-Fusion Foundation Models

Chameleon Team^{1,*}

¹FAIR at Meta

*See Contributions section for full author list.

We present Chameleon, a family of early-fusion token-based mixed-modal models capable of understanding and generating images and text in any arbitrary sequence. We outline a stable training approach from inception, an alignment recipe, and an architectural parameterization tailored for the early-fusion, token-based, mixed-modal setting. The models are evaluated on a comprehensive range of tasks, including visual question answering, image captioning, text generation, image generation, and long-form mixed modal generation. Chameleon demonstrates broad and general capabilities, including state-of-the-art performance in image captioning tasks, outperforms Llama-2 in text-only tasks while being competitive with models such as Mixtral 8x7B and Gemini-Pro, and performs non-trivial image generation, all in a single model. It also matches or exceeds the performance of much larger models, including Gemini Pro and GPT-4V, according to human judgments on a new long-form mixed-modal generation evaluation, where either the prompt or outputs contain mixed sequences of both images and text. Chameleon marks a significant step forward in a unified modeling of full multimodal documents.

Date: May 17, 2024



<https://github.com/facebookresearch/chameleon>

Unified image and text generation: Chameleon

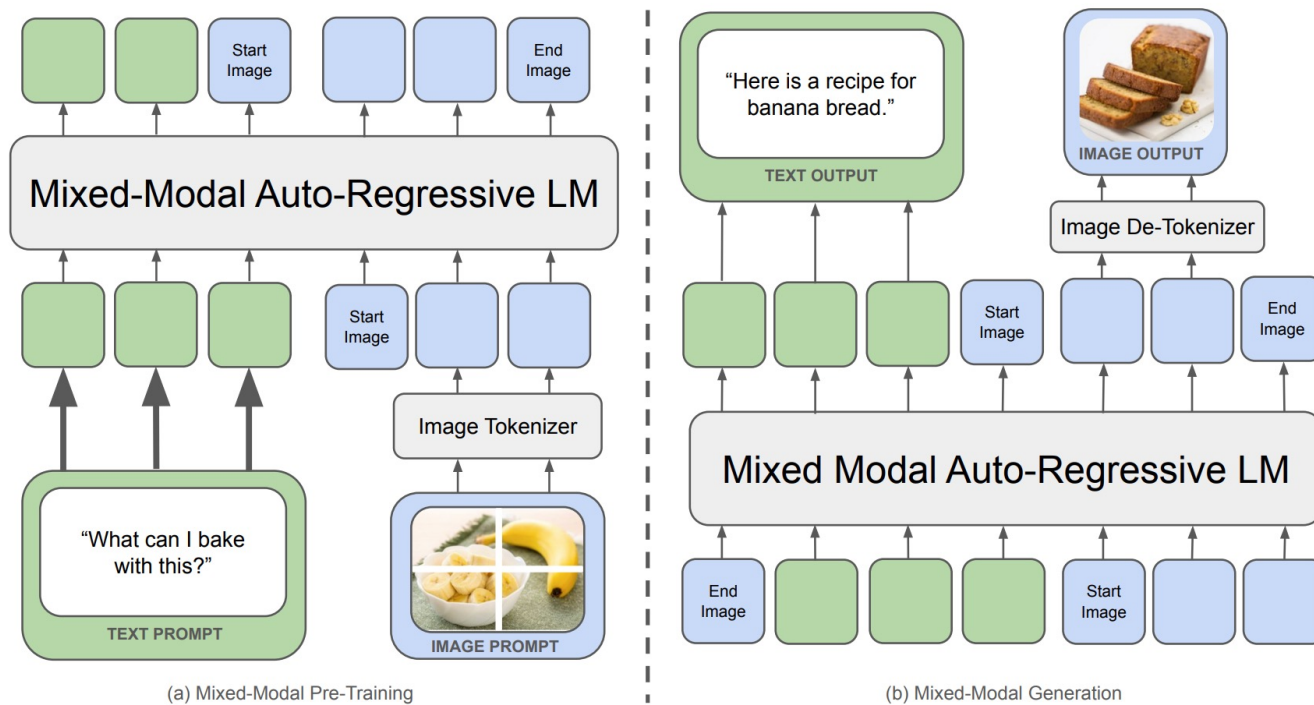
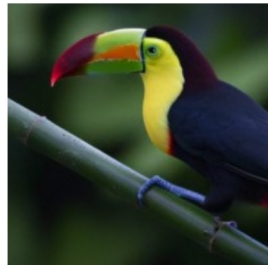


Figure 1 Chameleon represents all modalities — images, text, and code, as discrete tokens and uses a uniform transformer-based architecture that is trained from scratch in an end-to-end fashion on $\sim 10T$ tokens of interleaved mixed-modal data. As a result, Chameleon can both reason over, as well as generate, arbitrary mixed-modal documents. Text tokens are represented in green and image tokens are represented in blue.

Unified image and text generation: Chameleon

Prompt: I'm bored. Could you show me some cool, quirky-looking birds? Also tell me about them in 2-3 sentences.



Response: Absolutely! Here are three birds with some unique features that you might find interesting:

Keel-Billed Toucan The Keel-Billed Toucan is a brightly colored bird native to South American rainforests and woodlands. Its most distinctive feature is its colorful beak, which can grow up to seven inches long and has a distinctive, curved shape. The beak is used for plucking fruit from trees and for social displays. Despite its large size, the Keel-Billed Toucan is a skilled flyer and can often be seen flitting through the treetops. Here is an image of this vibrant bird: ****

Puffin Puffins are small, aquatic birds that are found in the northern hemisphere. They have distinctive black and white feathers, as well as a brightly colored beak that is used for attracting mates and defending their territory. Puffins are known for their distinctive "puffin" call, which sounds like a loud, guttural chuckle. They feed on fish, which they catch by diving into the water from the air. Puffins are colonial birds, and they breed in large colonies on rocky cliffs and islands. Here is an image of this adorable bird: ****



Golden Pheasant The Golden Pheasant is a brightly colored bird native to the forests of China. It is a member of the pheasant family and is known for its striking plumage, which features a golden-yellow body, red face, and green tail feathers. The Golden Pheasant is a ground-dwelling bird, and it feeds on a diet of insects, seeds, and fruits. It is a popular game bird, and is often hunted for its meat and feathers. In captivity, the Golden Pheasant is sometimes kept as an ornamental bird, due to its striking appearance. Here is an image of this beautiful bird: ****

I hope these images and descriptions have helped to brighten your day! If you have any other questions or requests, please don't hesitate to ask.

Diffusion for text generation

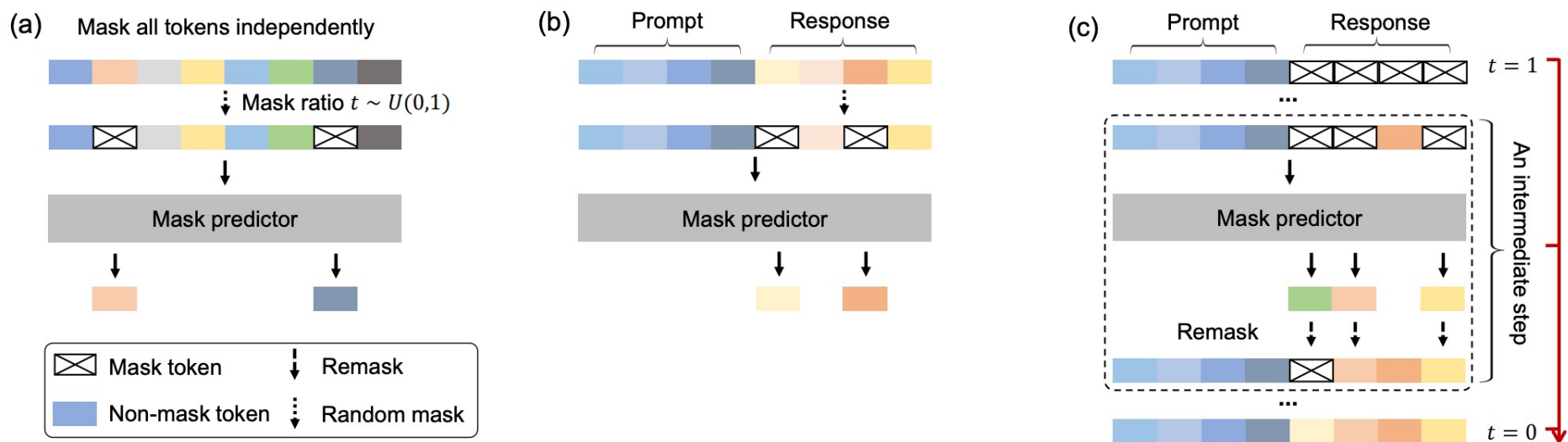


Figure 2. A Conceptual Overview of LLaDA. (a) Pre-training. LLaDA is trained on text with random masks applied independently to all tokens at the same ratio $t \sim U[0, 1]$. (b) SFT. Only response tokens are possibly masked. (c) Sampling. LLaDA simulates a diffusion process from $t = 1$ (fully masked) to $t = 0$ (unmasked), predicting all masks simultaneously at each step with flexible remask strategies.

Diffusion for text generation

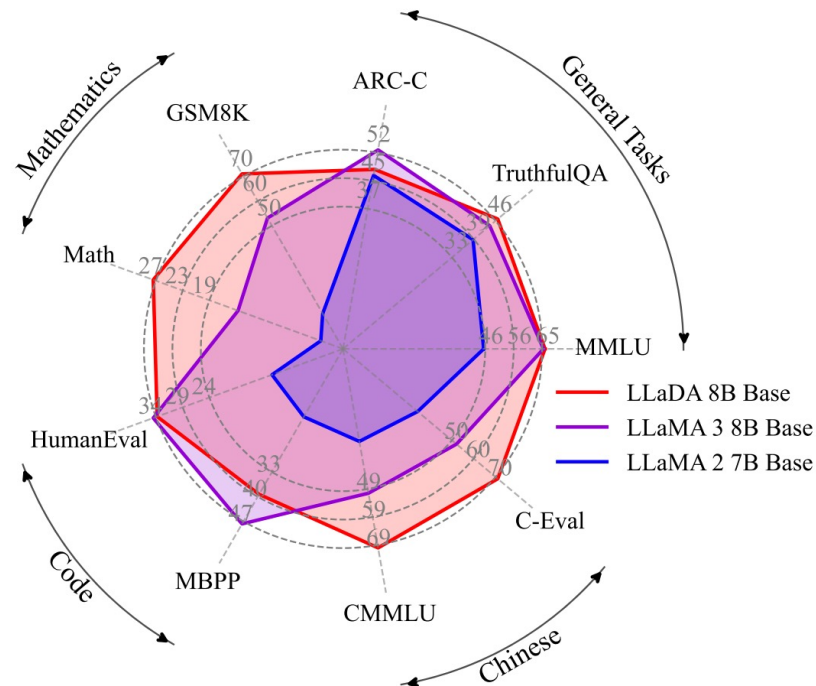


Figure 1. Zero/Few-Shot Benchmarks. We scale LLaDA to an unprecedented size of 8B parameters from scratch, achieving competitive performance with strong LLMs (Dubey et al., 2024).

S. Nie et al. [Large Language Diffusion Models](#). arXiv 2025

The era of experience?

The Era of Human Data

Artificial intelligence (AI) has made remarkable strides over recent years by training on massive amounts of human-generated data and fine-tuning with expert human examples and preferences. This approach is exemplified by large language models (LLMs) that have achieved a sweeping level of generality. A single LLM can now perform tasks spanning from writing poetry and solving physics problems to diagnosing medical issues and summarising legal documents.

However, while imitating humans is enough to reproduce many human capabilities to a competent level, this approach in isolation has not and likely cannot achieve superhuman intelligence across many important topics and tasks. In key domains such as mathematics, coding, and science, the knowledge extracted from human data is rapidly approaching a limit. The majority of high-quality data sources - those that can actually improve a strong agent's performance - have either already been, or soon will be consumed. The pace of progress driven solely by supervised learning from human data is demonstrably slowing, signalling the need for a new approach. Furthermore, valuable new insights, such as new theorems, technologies or scientific breakthroughs, lie beyond the current boundaries of human understanding and cannot be captured by existing human data.

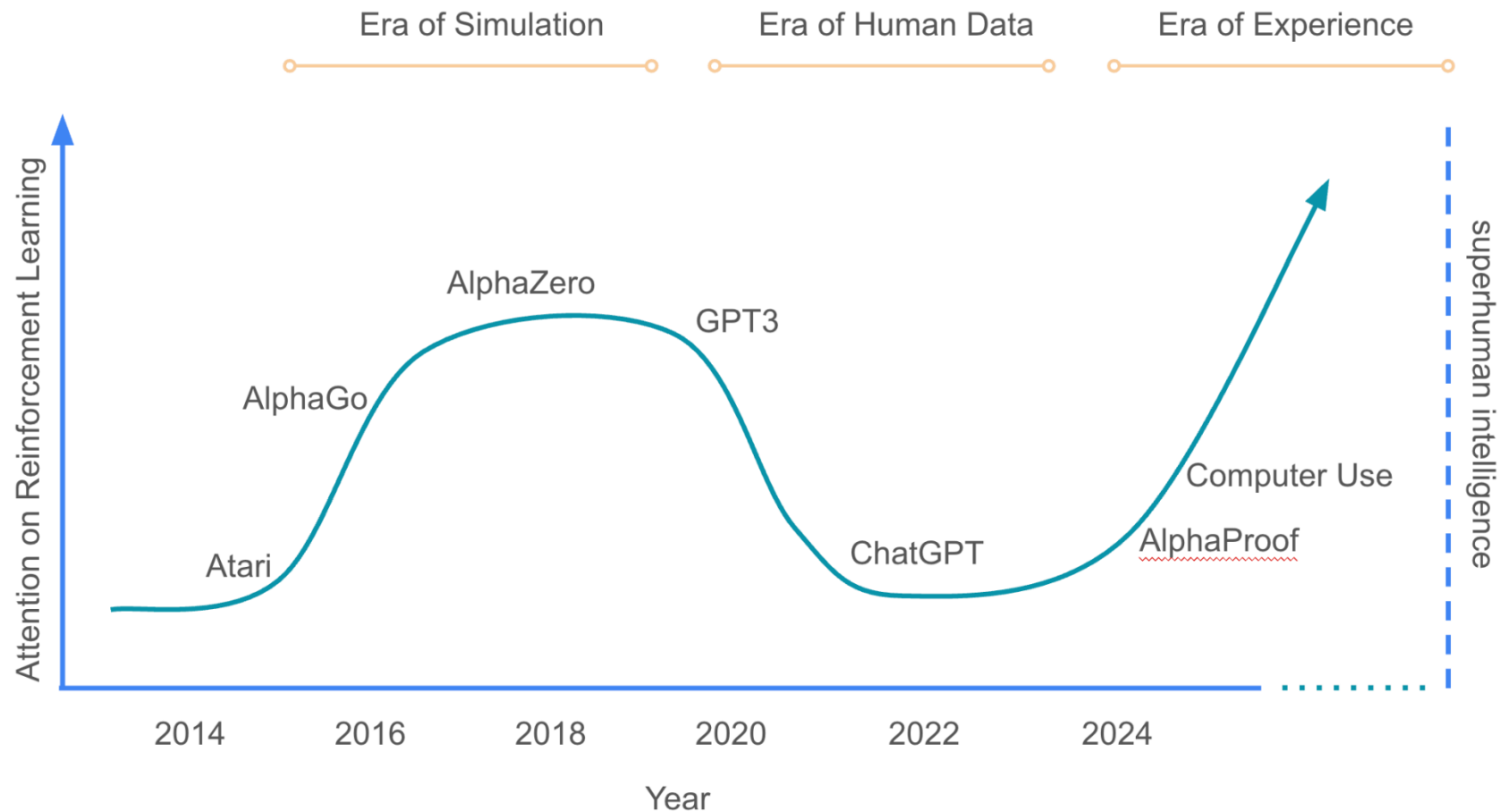
The Era of Experience

To progress significantly further, a new source of data is required. This data must be generated in a way that continually improves as the agent becomes stronger; any static procedure for synthetically generating data will quickly become outstripped. This can be achieved by allowing agents to learn continually from their own *experience*, i.e., data that is generated by the agent interacting with its environment. AI is at the cusp of a new period in which experience will become the dominant medium of improvement and ultimately dwarf the scale of human data used in today's systems.

[D. Silver](#) and [R. Sutton](#). [Welcome to the Era of Experience](#). Preprint, 2025

See also: R. Sutton, [The Bitter Lesson](#)

The era of experience?



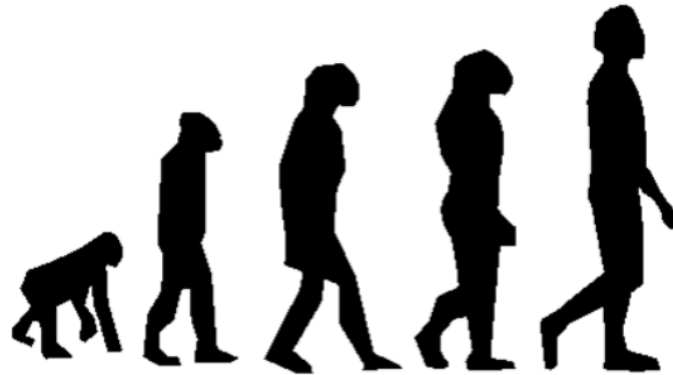
The bitter lesson

We have to learn the bitter lesson that building in how we think we think does not work in the long run. The bitter lesson is based on the historical observations that 1) AI researchers have often tried to build knowledge into their agents, 2) this always helps in the short term, and is personally satisfying to the researcher, but 3) in the long run it plateaus and even inhibits further progress, and 4) breakthrough progress eventually arrives by an opposing approach based on scaling computation by search and learning. The eventual success is tinged with bitterness, and often incompletely digested, because it is success over a favored, human-centric approach.

One thing that should be learned from the bitter lesson is the great power of general purpose methods, of methods that continue to scale with increased computation even as the available computation becomes very great. The two methods that seem to scale arbitrarily in this way are *search* and *learning*.

The second general point to be learned from the bitter lesson is that the actual contents of minds are tremendously, irredeemably complex; we should stop trying to find simple ways to think about the contents of minds, such as simple ways to think about space, objects, multiple agents, or symmetries. All these are part of the arbitrary, intrinsically-complex, outside world. They are not what should be built in, as their complexity is endless; instead we should build in only the meta-methods that can find and capture this arbitrary complexity. Essential to these methods is that they can find good approximations, but the search for them should be by our methods, not by us. We want AI agents that can discover like we can, not which contain what we have discovered. Building in our discoveries only makes it harder to see how the discovering process can be done.

True Intelligence Must Emerge



*“AI is not when computer can write poetry,
AI is when computer will **want** to write poetry”*
-- young soviet physicist in the 1980s

Source: A. Efros